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### DISPLAY SYSTEM

#### Abstract

A display device includes a first display device that projects a first virtual image ahead of a user in a vehicle; and a second display device that projects a second virtual image ahead of the user and below the first virtual image. A lower end of the first virtual image and an upper end of the second virtual image are located within a circle whose diameter is a length of the second virtual image in a side view.

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## Background/Summary

### CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application is based on and claims priority of Japanese Patent Application No. 2024-031098 filed on Mar. 1, 2024, Japanese Patent Application No. 2024-031110 filed on Mar. 1, 2024, and Japanese Patent Application No. 2024-154389 filed on Sep. 9, 2024.

### FIELD

[0002] The present disclosure relates to a display system.

### BACKGROUND

[0003] Patent Literature (PTL) 1 discloses a display system that projects a first virtual image and a second virtual image onto positions that are approximately the same distances ahead of the user such that the user can easily view the first virtual image and the second virtual image.

### CITATION LIST

#### Patent Literature

[0004] PTL 1: Japanese Unexamined Patent Application Publication No. 2015-146012

### SUMMARY

[0005] However, the foregoing display system can be improved upon.

[0006] In view of this, the present disclosure provides a display system capable of improving upon the above related art.

[0007] The display system according to an aspect of the present disclosure includes: a first display device that projects a first virtual image ahead of a user in a vehicle; and a second display device that projects a second virtual image ahead of the user and below the first virtual image, wherein a lower end of the first virtual image and an upper end of the second virtual image are located within a circle whose diameter is a length of the second virtual image in a side view.

[0008] The display system according to the present disclosure is capable of improving upon the above related art.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

[0009] These and other advantages and features of the present disclosure will become apparent from the following description thereof taken in conjunction with the accompanying drawings that illustrate a specific embodiment of the present disclosure.

[0010] FIG. 1 is a schematic diagram showing a display system according to Embodiment 1 disposed in a vehicle.

[0011] FIG. 2 is a schematic cross-sectional diagram showing a first display device according to Embodiment 1.

[0012] FIG. 3 is a schematic cross-sectional diagram showing a second display device according to Embodiment 1.

[0013] FIG. 4 is a schematic diagram showing a display system according to Embodiment 2.

[0014] FIG. 5 is a schematic cross-sectional diagram showing a third display device according to Embodiment 2.

[0015] FIG. 6 is a schematic diagram showing a third display device according to Variation 1 of Embodiment 2.

[0016] FIG. **7** is a schematic diagram showing third display devices according to Variation 2 of Embodiment 2.

[0017] FIG. **8** is a schematic diagram showing a second display device according to Embodiment 3.

[0018] FIG. **9** is an explanatory diagram showing an example of the display performed by a display element according to Embodiment 3.

[0019] FIG. **10** is a schematic diagram showing a first display device according to Embodiment 3.

[0020] FIG. **11** is a schematic diagram showing a second display device according to Embodiment 4.

[0021] FIG. **12** is a schematic diagram showing a first display device according to Embodiment 5.

[0022] FIG. **13** is a schematic diagram showing a second display device according to Embodiment 5.

[0023] FIG. **14A** is an explanatory diagram showing an example of the layout of a first virtual image and a second virtual image according to Embodiment 6.

[0024] FIG. **14B** is an explanatory diagram showing an example of the layout of the first virtual image and the second virtual image according to Embodiment 6.

[0025] FIG. **14C** is an explanatory diagram showing an example of the layout of the first virtual image and the second virtual image according to Embodiment 6.

[0026] FIG. **15** is a schematic diagram showing the first display device corresponding to Example Layout 3 of Embodiment 6.

[0027] FIG. **16** is a schematic diagram showing a first display device according to Embodiment 7.

[0028] FIG. **17** is a schematic diagram showing a first display device and a second display device according to Embodiment 8.

[0029] FIG. **18** is an explanatory diagram showing the first virtual image and the second virtual image in reference positions in a display system according to Embodiment 8.

[0030] FIG. **19** is an explanatory diagram showing the first virtual image and the second virtual image in the reference positions seen from a driver in Embodiment 8.

[0031] FIG. **20** is an explanatory diagram showing a state in which a large driver takes over driving, and the first display position and the second display position have been adjusted in Embodiment 8.

[0032] FIG. **21A** is an explanatory diagram showing the first virtual image and the second virtual image only after the first display position has been adjusted in Embodiment 8.

[0033] FIG. **21B** is an explanatory diagram showing the first virtual image and the second virtual image in the case where the first display position has been adjusted and the coordinates of the first virtual image have been adjusted in Embodiment 8.

[0034] FIG. **22** is a schematic diagram showing a display system according to Embodiment 9 disposed in a vehicle.

[0035] FIG. **23** is an explanatory diagram showing an example of displaying items of notification information according to Embodiment 9.

[0036] FIG. **24** is a schematic diagram showing a display system according to Embodiment 10.

[0037] FIG. **25** is a schematic diagram showing a display system according to Embodiment 11.

[0038] FIG. **26** is a schematic diagram showing a display system according to Embodiment 12.

[0039] FIG. **27** is an explanatory diagram showing an example of displaying items of notification information according to Embodiment 13.

[0040] FIG. **28** is an explanatory diagram showing another example of displaying a mark according to Embodiment 13.

[0041] FIG. **29** is an explanatory diagram showing another example of displaying items of notification information according to Embodiment 13.

[0042] FIG. **30** is an explanatory diagram showing an example of displaying items of notification information according to Embodiment 14.

[0043] FIG. **31** is an explanatory diagram showing an example of displaying items of notification

information according to Embodiment 15.

[0044] FIG. **32** is an explanatory diagram showing another example of displaying items of notification information according to Embodiment 15.

[0045] FIG. **33** is a schematic diagram showing a display system according to Embodiment 16.

[0046] FIG. **34** is an explanatory diagram showing an example of the display performed by a second display device according to Embodiment 16.

[0047] FIG. **35** is an explanatory diagram showing an example of displaying video according to Embodiment 16.

[0048] FIG. **36** is an explanatory diagram showing another example of the display according to Embodiment 16.

[0049] FIG. **37** is a schematic diagram showing a display system according to Embodiment 17.

[0050] FIG. **38** is a schematic diagram showing a display system according to Embodiment 18.

[0051] FIG. **39** is an explanatory diagram showing an example of displaying notification information according to Embodiment 18.

[0052] FIG. **40** is an explanatory diagram showing another example of displaying notification information according to Embodiment 18.

[0053] FIG. **41** is an explanatory diagram showing another example of displaying notification information according to Embodiment 18.

[0054] FIG. **42** is a schematic diagram showing another example of the display system according to Embodiment 18.

[0055] FIG. **43** is a schematic diagram showing a display system according to Embodiment 19.

[0056] FIG. **44** is a schematic diagram showing a display system according to Embodiment 20.

[0057] FIG. **45** is a schematic diagram showing the display system according to Embodiment 20.

[0058] FIG. **46** is a schematic diagram showing the display system according to Embodiment 20.

## DESCRIPTION OF EMBODIMENTS

### Underlying Knowledge Forming Basis of the Present Disclosure

[0059] In recent years, there has been a demand for further alleviating the burden on a user in viewing different virtual images. Stated differently, the present disclosure provides a display system capable of alleviating the burden on a user in viewing different virtual images.

[0060] The display system according to an aspect of the present disclosure includes: a first display device that projects a first virtual image ahead of a user in a vehicle; and a second display device that projects a second virtual image ahead of the user and below the first virtual image, wherein a lower end of the first virtual image and an upper end of the second virtual image are located within a circle whose diameter is a length of the second virtual image in a side view.

[0061] With this, the lower end of the first virtual image and the upper end of the second virtual image are located within the circle whose diameter is the length of the second virtual image in a side view. This enables the lower end of the first virtual image and the upper end of the second virtual image to be located in proximity to each other in the up-down direction. This reduces each of the gap in the up-down direction and the gap in the horizontal direction between the lower end of the first virtual image and the upper end of the second virtual image. It is thus possible to alleviate the burden on the user caused by moving his/her viewpoint and focus point, switching between the first virtual image and the second virtual image, in viewing these virtual images.

[0062] Also, a difference between a first angle of depression and a second angle of depression may be  $12^\circ$  or less, the first angle of depression being an angle formed by the lower end of the first virtual image with respect to a viewpoint of the user, the second angle of depression being an angle formed by the upper end of the second virtual image with respect to the viewpoint of the user.

[0063] With this, the difference between the first angle of depression formed by the lower end of the first virtual image with respect to the viewpoint of the user and the second angle of depression formed by the upper end of the second virtual image with respect to the viewpoint of the user is  $12^\circ$  or less. This further reduces the gap in the up-down direction between the lower end of the first

virtual image and the upper end of the second virtual image. It is thus possible to further alleviate the burden on the user caused by moving his/her viewpoint and focus point, switching between the first virtual image and the second virtual image, in viewing these virtual images.

[0064] Also, the display system may include a third display device that projects a third virtual image ahead of the user, and the third virtual image may be located at a viewing distance of 0.25 diopters or less with respect to a viewing distance of the second virtual image.

[0065] With this, the third virtual image is located at the viewing distance of 0.25 diopters or less with respect to the viewing distance of the second virtual image. This reduces the gap in the horizontal direction between the second virtual image and the third virtual image. It is thus possible to alleviate the burden on the user caused by moving his/her viewpoint and focus point, switching between the second virtual image and the third virtual image, in viewing these virtual images.

[0066] Also, at least one of the first virtual image projected by the first display device or the second virtual image projected by the second display device may be displayed in a plurality of layers.

[0067] With this, at least one of the first virtual image or the second virtual image is displayed in a plurality of layers. This enables different contents to be displayed on a layer-by-layer basis, thereby enabling more diverse content representation.

[0068] Also, the second display device may include: a display element that emits video light for forming the second virtual image; and an optical system for projecting the video light from the display element as the second virtual image, and the optical system may include a concave mirror that reflects the video light at a final stage.

[0069] With this, the second virtual image is formed by means of the concave mirror reflecting and converging the video light at the final stage. It is thus possible to easily secure the viewing distance of the second virtual image.

[0070] Also, at least one of the first display device or the second display device may include an adjuster for adjusting the viewing distance of the first virtual image and the viewing distance of the second virtual image.

[0071] With this, at least one of the first display device or the second display device includes the adjuster for adjusting the viewing distance of the first virtual image and the viewing distance of the second virtual image. This enables the adjustment of the viewing distance of at least one of the first virtual image or the second virtual image. With this, it is possible to adjust the viewing distance of at least one of the first virtual image or the second virtual image by means of the adjuster, in accordance with various conditions. This enables more diverse content representation.

[0072] Also, the display system may include a controller that controls a display content of the first display device and a display content of the second display device.

[0073] With this, the controller controls the display content of the first display device and the display content of the second display device. It is thus possible for the controller to collectively control the display content of the first display device (display content of the first virtual image) and the display content of the second display device (display content of the second virtual image). This facilitates the content coordination between the first virtual image and the second virtual image.

[0074] Also, at least one of the first display device or the second display device may include an optical system that includes a hologram element.

[0075] With this, at least one of the first display device or the second display device includes the optical system that includes the hologram element. It is thus possible for at least one of the first display device or the second display device to have a low-profile optical system.

[0076] Also, the display system may include a first position adjuster that adjusts a first display position in an up-down direction of the first virtual image; a second position adjuster that adjusts a second display position in an up-down direction of the second virtual image; and a coordinate changer that changes coordinates of the first virtual image within a displayable range of the first virtual image, based on the first display position and the second display position that have been adjusted.

[0077] With this, the display system includes the first position adjuster, the second position adjuster, and the coordinate changer. Thus, even when the viewpoint position of the driver has changed, it is possible to also adjust the coordinates of the first virtual image within the displayable range, while adjusting the first display position of the first virtual image and the second display position of the second virtual image. This enables the first virtual image and the second virtual image to be suitably displayed also for a different driver.

[0078] Also, the first display device may include a first reflector that reflects video light for forming the first virtual image, the second display device may include a second reflector that reflects video light for forming the second virtual image, the first position adjuster may adjust at least one of an orientation or a position of the first reflector to adjust the first display position of the first virtual image, and the second position adjuster may adjust at least one of an orientation or a position of the second reflector to adjust the second display position of the second virtual image.

[0079] With this, the first position adjuster adjusts the first display position of the first virtual image by adjusting at least one of the orientation or the position of the first reflector. It is thus possible to adjust the first display position using a simple structure. Meanwhile, the second position adjuster adjusts the second display position of the second virtual image by adjusting at least one of the orientation or the position of the second reflector. It is thus possible to adjust the second display position using a simple structure. These enable the first virtual image and the second virtual image to be suitably displayed also for a different driver.

[0080] Also, the display system may include a head imager that captures an image of a head of the user; an estimator that estimates a viewpoint position of the user, based on the image captured by the head imager; and an adjustment controller that controls the first position adjuster and the second position adjuster, based on the viewpoint position of the user estimated.

[0081] With this, the estimator estimates the viewpoint position of the user, on the basis of the image captured by the head imager, and the adjustment controller controls the first position adjuster and the second position adjuster, on the basis of such estimated viewpoint position. It is thus possible to automatically adjust the first display position and the second display position.

[0082] Also, the display system may include a first imager that captures images of surroundings of the vehicle; and a controller that controls the first display device, the second display device, and the first imager. When the controller causes each of the first display device and the second display device to display notification information to be provided to the user, the controller may adjust, between the first display device and the second display device, at least one of luminance, a size, or chromaticity of the notification information to be displayed together with vehicle surrounding information obtained from the first imager.

[0083] With this, when the notification information is displayed by each of the first display device and the second display device, at least one of the luminance, the size, or the chromaticity of the notification information to be displayed together with the vehicle surrounding information obtained from the first imager is adjusted between the first display device and the second display device. It is thus possible to display the notification information displayed by the first display device and the notification information displayed by the second display device in a visually uniform (consistent) manner. This reduces the user's sense of discomfort that occurs when such user views the notification information in the first virtual image and the notification information in the second virtual image.

[0084] Also, the first imager may capture the images of an area ahead of the vehicle, and the controller may calculate luminance of the first virtual image in a state of being overlaid on a background, based on the vehicle surrounding information, and adjust the luminance of the notification information between the first display device and the second display device, based on a calculation result.

[0085] With this, the luminance of the first virtual image in the state of being overlaid on the background is calculated on the basis of the vehicle surrounding information. Then, on the basis of

the calculation result, the luminance of each of the items of notification information is adjusted between the first display device and the second display device. With this, it is possible to reflect, on the luminance of each of the items of notification information, the influence of the background exerted on the luminance of the first virtual image. This enables information notification that creates a lesser sense of discomfort.

[0086] Also, the controller may adjust the luminance of the notification information between the first display device and the second display device to cause the luminance of the notification information in the first virtual image and the luminance of the notification information in the second virtual image to be approximately same.

[0087] With this, the luminance of the notification information in the first virtual image and the luminance of the notification information in the second virtual image are adjusted to be approximately the same. It is thus possible to reduce the difference in the luminance between the notification information in the first virtual image and the notification information in the second virtual image. This further reduces the user's sense of discomfort that occurs when such user views the notification information in the first virtual image and the notification information in the second virtual image.

[0088] The controller may adjust the luminance of the notification information between the first display device and the second display device to cause the luminance of the notification information in the first virtual image to be higher than the luminance of the notification information in the second virtual image.

[0089] With this, the luminance of the notification information in the first virtual image is adjusted to be higher than the luminance of the notification information in the second virtual image. It is thus possible to make the notification information in the first virtual image, which is located above the second virtual image, more stand out. This enables the user to more easily recognize the notification information in the first virtual image.

[0090] The controller may adjust the luminance of the notification information between the first display device and the second display device to cause the luminance of the notification information in the first virtual image to be lower than the luminance of the notification information in the second virtual image.

[0091] With this, the luminance of the notification information in the first virtual image is adjusted to be lower than the luminance of the notification information in the second virtual image. It is thus possible to make the notification information in the second virtual image, which is displayed below the first virtual image and thus directs the line of sight of the user downward, more stand out.

[0092] The display system may further include a second imager that captures an image of surroundings of the user, and the controller may adjust the luminance of the notification information between the first display device and the second display device, on the basis of user surrounding information obtained from the second imager.

[0093] With this, it is possible to obtain the user surrounding information by means of the second imager. This enables to have an accurate grasp of the viewpoint position of the user, on the basis of such user surrounding information. With this, it is possible to have an accurate grasp of the luminance viewed by the user from the viewpoint position, thereby reflecting such luminance on the luminance of each of the items of notification information. This thus enables information notification that creates a lesser sense of discomfort.

[0094] The display system may further include an illuminance sensor that detects an illuminance of an area in the vicinity of the second display device, and the controller may adjust the luminance the notification information between the first display device and the second display device, based on a detection result of the illuminance sensor.

[0095] With this, the luminance of the notification information is adjusted on the basis of the detection result of the illuminance sensor. It is thus possible to reflect the illuminance of the area in the vicinity of the second display device on the luminance of each of the items of notification

information. This thus enables information notification that creates a lesser sense of discomfort.

[0096] The display system may further include a second imager that captures an image of surroundings of the user, and the controller may adjust the size of the notification information between the first display device and the second display device, on the basis of user surrounding information obtained from the second imager.

[0097] With this, it is possible to obtain the user surrounding information by means of the second imager. This enables to have an accurate grasp of the viewpoint position of the user, on the basis of such user surrounding information. With this, it is possible to accurately adjust the size of each of the items of notification information viewed by the user from the viewpoint position, thereby enabling information notification that creates a lesser sense of discomfort.

[0098] The first imager may capture the images of an area ahead of the vehicle, and the controller may calculate chromaticity of the first virtual image in a state of being overlaid on a background, based on the vehicle surrounding information, and adjust, between the first display device and the second display device, the chromaticity of the notification information in the first virtual image and the chromaticity of the notification information in the second virtual image to be approximately same, based on a calculation result.

[0099] With this, the chromaticity of the first virtual image in the state of being overlaid on the background is calculated on the basis of the vehicle surrounding information. Then, on the basis of the calculation result, the chromaticity of the notification information in the first virtual image and the chromaticity of the notification information in the second virtual image are adjusted to be approximately the same. With this, it is possible to adjust the chromaticity of the notification information in the first virtual image and the chromaticity of the notification information in the second virtual image to be approximately the same, in consideration of the influence of the background exerted on the first virtual image. This enables information notification that creates a lesser sense of discomfort.

[0100] Also, the controller may adjust the chromaticity of the notification information in the first virtual image, with the chromaticity of the notification information in the second virtual image kept constant.

[0101] With this, the chromaticity of the notification information in the first virtual image is adjusted, with the chromaticity of the notification information in the second virtual image kept constant. It is enables the notification information in the first virtual image to be more easily viewed.

[0102] The controller may adjust the chromaticity of the notification information in the second virtual image to be approximately the same as the chromaticity of the notification information in the first virtual image, with the chromaticity of the notification information in the first virtual image kept constant.

[0103] With this, the chromaticity of the notification information in the second virtual image is adjusted to be approximately the same as the chromaticity of the notification information in the first virtual image, with the chromaticity of the notification information in the first virtual image kept constant. This enables the chromaticity of the notification information in the first virtual image to be less likely to change, thus alleviating the burden on the user when looking ahead.

[0104] The display system may further include a second imager that captures an image of surroundings of the user, and the controller may adjust the chromaticity of the notification information between the first display device and the second display device, based on user surrounding information obtained from the second imager.

[0105] With this, it is possible to obtain the user surrounding information by means of the second imager. This enables to have an accurate grasp of the viewpoint position of the user, on the basis of such user surrounding information. With this, it is possible to have an accurate grasp of the chromaticity viewed by the user from the viewpoint position, thereby reflecting such chromaticity on the chromaticity of each of the items of notification information. This thus enables information



notification that creates a lesser sense of discomfort.

[0106] Also, the controller may cause the notification information to be displayed, based on a display reference point that is set inside a first display area of the first display device. When a target object for which the notification information is provided is present inside the first display area, the controller may set the target object as the display reference point, and when the target object for which the notification information is provided is present outside the first display area, the controller may set the display reference point on a center line in a width direction of the first display area.

[0107] With this, when the target object for which each of the items of notification information is displayed is present inside the first display area, the items of notification information are displayed, with such target object serving as the display reference point. This enables the user to also recognize the target object simply by viewing the items of notification information. Meanwhile, when the target object for which each of the items of notification information is displayed is present outside the first display area, the notification information is displayed on the display reference point that is set on the center line in the width direction of the first display area. Stated differently, since the notification information is located on the center line in the first display area, when the target object is not present inside the first display area, it is possible for the user to more reliably view the notification information.

[0108] Also, the controller may cause the notification information displayed by the first display device and the notification information displayed by the second display device to be located on a straight line that includes the display reference point.

[0109] With this, the notification information displayed by the first display device and the notification information displayed by the second display device are located on the straight line that includes the display reference point. It is thus possible for the user to linearly move the line of sight to each notification information. This alleviates the burden on the user caused by moving the line of sight.

[0110] Also, the controller may cause the notification information displayed by the first display device and the notification information displayed by the second display device to be displayed in a continuous manner.

[0111] With this, the notification information displayed by the first display device and the notification information displayed by the second display device are displayed in a continuous manner. It enables the user to more easily recognize the positional relationship between the items of notification information.

[0112] Also, the controller may cause the notification information displayed by the first display device and the notification information displayed by the second display device to be displayed in a manner that each of the notification information displayed by the first display device and the notification information displayed by the second display device moves from the first display area of the first display device toward a second display area of the second display device.

[0113] With this, the notification information displayed by the first display device and the notification information displayed by the second display device are displayed in a manner that these items of notification information move from the first display area to the second display area. This makes it easy to guide the line of sight of the user.

[0114] Also, the controller may cause the second display device to display video captured by the first imager.

[0115] With this, the video captured by the first imager is displayed by the second display device. It is thus possible to display the video captured by the first imager and the notification information displayed by the second display device in an overlaid manner. This enables the notification information to be displayed on a clear video, thereby enabling content representation that is easy for the user to understand.

[0116] Also, the display device may include a side imager that captures video of surroundings of a lateral side of the vehicle, and the controller may cause the second display device to display a

pseudo image overlaid on the video captured by the side imager, the pseudo image being an image that mimics an interior of the vehicle as a blind spot of the user.

[0117] With this, the controller causes the second display device to display the pseudo image, which mimics the interior of the vehicle as a blind spot portion of the user, in a manner that such pseudo image is overlaid on the video captured by the side imager. It is thus possible for the user to view the pseudo image displayed by the second display device and the video captured by the side imager. This enables the user to recognize the situations on the lateral sides of the vehicle and the blind spot, and thus have a grasp of the surrounding situation of the vehicle.

[0118] The display system may include a tilt sensor that obtains a tilt of the vehicle, and the controller may correct the video captured by the side imager, based on a detection result of the tilt sensor, to cause the second display device to display the video corrected.

[0119] With this, the second display device displays the video that has been corrected on the basis of the detection result of the tilt sensor. It is thus possible for the user to have a more accurate grasp of the surrounding situation of the vehicle.

[0120] Also, the display device may include a generator that generates a sound or a vibration, and the controller may cause the generator to generate the sound or the vibration, based on a display timing of the notification information.

[0121] With this, the generator generates the sound or the vibration on the basis of the display timing of the items of notification information. It is thus possible for the user to more reliably have a grasp of the notification information in response to the sound or the vibration.

[0122] The first display device may include: an augmented reality head-up display that projects one portion of the first virtual image; and a head-up display that projects another portion of the first virtual image.

[0123] With this, the first virtual image is formed by the augmented reality head-up display and the head-up display. It is thus possible to perform more diverse representation of the first virtual image.

## EMBODIMENTS

[0124] Hereinafter, certain exemplary embodiments are described in greater detail with reference to the accompanying Drawings. Each of the exemplary embodiments described below shows a general or specific example. The numerical values, shapes, materials, elements, the arrangement and connection of the elements, steps, the processing order of the steps etc. shown in the following exemplary embodiments are mere examples, and therefore do not limit the scope of the present disclosure. Therefore, among the elements in the following exemplary embodiments, those not recited in any one of the independent claims are described as optional elements.

[0125] In the following embodiments, expressions that indicate relative orientations of two directions, such as “parallel” and “orthogonal”, are used in some cases, but these expressions also mean the case where the two directions are not in such orientations in a strict sense. When the two directions are described as being parallel, for example, such expression not only means that the two directions are perfectly parallel, but also that the two directions are substantially parallel, unless otherwise stated. Stated differently, the expression also means that an error on the order of a few percent, for example, is allowed. Also, in the following embodiments, optical paths illustrated in the drawings are intended to show how the optical paths are interpreted in principle, and thus do not necessarily show the actual optical paths.

### Embodiment 1

[0126] FIG. 1 is a schematic diagram showing display system **10** according to Embodiment 1 disposed in vehicle **1**. In FIG. 1, vehicle **1** is shown in cross-section.

[0127] As shown in FIG. 1, display system **10** includes first display device **100**, second display device **200**, and controller **500**. First display device **100** and second display device **200** are disposed, for example, on the dashboard of vehicle **1**. First display device **100** and second display device **200** display vehicle information related to vehicle **1**, for example, as first virtual image **101** and second virtual image **201**. Examples of the vehicle information include the vehicle speed of

vehicle **1**, the number of revolutions of the engine, a detection result on an object that is present in the proximity of vehicle **1**, navigation information from the current position of vehicle **1** to the destination, image information captured by a camera that captures images of the rear and surroundings of vehicle **1**, etc. Note that FIG. **1** shows an example case where first display device **100** and second display device **200** are disposed on the dashboard, but the position where first display device **100** and second display device **200** is disposed is not limited to this; first display device **100** and second display device **200** may be disposed, for example, in the vicinity of the upper end of windshield **2** or in the center console.

[First Display Device]

[0128] First display device **100** is an augmented reality head-up display (AR-HUD). First display device **100** projects light onto region D1 in windshield (windscreen) **2** serving as a display medium. The projected light is reflected by windshield **2**. Such reflected light travels toward the eyes of the driver seated in the driver's seat, who is the user of first display device **100**. The driver perceives the reflected light that has entered his/her eyes as first virtual image **101** that is seen on the opposite side across windshield **2** (outside the vehicle), against the actual objects seen across windshield **2** serving the background.

[0129] The following describes the configuration of first display device **100** with reference to FIG. **2**. FIG. **2** is a schematic cross-sectional diagram showing first display device **100** according to Embodiment 1.

[0130] As shown in FIG. **2**, first display device **100** includes housing **110**, cover **120**, display element **130**, first optical element **140**, and second optical element **150**. First optical element **140** and second optical element **150** are optical systems for projecting video light from display element **130** as first virtual image **101**.

[0131] Housing **110** is a box-shaped body formed of lightproof resin or metal. More specifically, housing **110** has an approximately prismatic shape, and includes opening **111** that is provided in an upper portion thereof. Opening **111** is closed by cover **120**. The internal space defined by housing **110** and cover **120** accommodates display element **130**, first optical element **140**, and second optical element **150**.

[0132] Cover **120** is, for example, a curved plate body formed of a translucent resin or glass. More specifically, cover **120** has a shape that is convex downward as a whole.

[0133] Display element **130** is, for example, a liquid-crystal panel. When irradiated with light from a light source not shown, display element **130** emits, to first optical element **140**, video light to serve as first virtual image **101**. Display element **130** may also be an organic electro-luminescence (EL) panel. Display element **130** has a rectangular shape in a plan view and is disposed at an angle relative to the horizontal plane.

[0134] First optical element **140** is an optical element that is disposed on the optical path of the video light from display element **130** and reflects the video light toward second optical element **150**. First optical element **140** is a convex mirror having a rectangular shape in a plan view. First optical element **140** is disposed at an angle relative to the vertical plane. The reflecting surface of first optical element **140** faces display element **130** and second optical element **150**. Stated differently, in first optical element **140**, the reflecting surface, which is the mirror surface of the convex mirror, faces inward of housing **110**, and the concave surface faces outward of housing **110**.

[0135] Second optical element **150** is an optical member that is disposed on the optical path of the video light which has passed through first optical element **140** and that reflects the video light reflected by first optical element **140** toward opening **111**. More specifically, second optical element **150** is a concave mirror having a rectangular shape in a plan view. Second optical element **150** faces the reflecting surface of first optical element **140** and is disposed at an angle relative to the vertical plane of housing **110**. The reflecting surface of second optical element **150** faces first optical element **140** and cover **120**. Stated differently, in second optical element **150**, the reflecting surface, which is the mirror surface of the concave mirror, faces inward of housing **110**, and the

convex surface faces outward of housing **110**. The video light reflected by second optical element **150** is projected onto windshield **2** via opening **111**. This reflection causes the video light to travel toward the eyes of the driver seated in the driver's seat to serve as first virtual image **101**. FIG. **1** shows the position of first virtual image **101** that is seen from the viewpoint of the driver. It is possible to set this position by adjusting the viewing distance of the video light from display element **130** of first display device **100**. The viewing distance is the distance from the viewpoint of the driver to the imaging position of a virtual image (e.g., first virtual image **101**). The viewpoint of the driver is, for example, the reference eyepoint. The reference eyepoint is "the point that represents the position of the eyes of the driver under normal driving conditions".

[Second Display Device]

[0136] As shown in FIG. **1**, second display device **200** projects the video light toward the driver. The driver perceives the video light that has entered his/her eyes as second virtual image **201** that appears in a distant position from opening **221** of second display device **200** (see FIG. **3**).

[0137] The following describes the configuration of second display device **200** with reference to FIG. **3**. FIG. **3** is a schematic cross-sectional diagram showing second display device **200** according to Embodiment 1. As shown in FIG. **3**, second display device **200** includes housing **220**, display element **230**, polarizing half mirror **240**, first reflecting mirror **250**, and second reflecting mirror **260**. Polarizing half mirror **240**, first reflecting mirror **250**, and second reflecting mirror **260** are optical systems for projecting the video light from display element **230** as second virtual image **201**.

[0138] Housing **220** is a box-shaped body formed of lightproof resin or metal. Opening **221** that faces rearward is provided at an upper end portion of the rear part of housing **220** (rightward direction in FIG. **3** is defined as the rear part and rearward). Video light to serve as second virtual image **201** is projected from opening **221**. The internal space defined by housing **220** accommodates display element **230**, polarizing half mirror **240**, first reflecting mirror **250**, and second reflecting mirror **260**.

[0139] Display element **230** is, for example, a liquid-crystal panel. When irradiated with light from a light source not shown, display element **230** emits, to polarizing half mirror **240**, the video light to serve as second virtual image **201**. Display element **230** may also be an organic EL panel. Display element **230** is disposed in an orientation in which the display surface faces rearward. Although not shown, a  $\lambda/4$  phase difference plate (hereafter, abbreviated as " $\lambda/4$  plate") is stacked on the display surface of display element **230**. The  $\lambda/4$  plate is a  $\lambda/4$  phase difference plate that causes a phase difference of  $1/4$  wavelength  $\lambda$  in light that has entered such  $\lambda/4$  plate. When the light outputted from the display surface is linear S-polarized light, for example, such light is converted into circularly polarized light by passing through the  $\lambda/4$  plate.

[0140] Polarizing half mirror **240** is configured to reflect P-polarized light and transmit S-polarized light. In polarizing half mirror **240**, a reflective polarizing plate is disposed on a glass substrate having a flat plate shape. In addition, a  $\lambda/4$  plate is stacked on the surface of polarizing half mirror **240**. Polarizing half mirror **240** is disposed in an orientation in which polarizing half mirror **240** faces display element **230** and first reflecting mirror **250**. The S-polarized video light outputted from display element **230** is converted into circularly polarized light by the  $\lambda/4$  plate stacked on display element **230**, and then travels toward polarizing half mirror **240**. Such circularly polarized video light is converted into P-polarized light by the  $\lambda/4$  plate stacked on polarizing half mirror **240**, and reflected by the reflective polarizing plate of polarizing half mirror **240**. The reflected P-polarized video light is converted into circularly polarized light by passing through the  $\lambda/4$  plate again. Thus, polarizing half mirror **240** is disposed in an orientation in which the video light that has entered as circularly polarized light is reflected by the  $\lambda/4$  plate and the reflective polarizing plate stacked on polarizing half mirror **240** toward first reflecting mirror **250** as circularly polarized light.

[0141] First reflecting mirror **250** is a concave mirror and disposed below polarizing half mirror

**240** in FIG. 3. First reflecting mirror **250** is disposed in an orientation in which the concave surface serving as the reflecting surface faces upward. The circularly polarized video light reflected by polarizing half mirror **240** is reflected by first reflecting mirror **250** as it is, that is, as circularly polarized light, and then travels toward polarizing half mirror **240** again. The video light that has entered polarizing half mirror **240** is converted into S-polarized light by the  $\lambda/4$  plate stacked on polarizing half mirror **240**, and then passes through the reflective polarizing plate of polarizing half mirror **240** to travel upward in FIG. 3.

[0142] Second reflecting mirror **260** is a flat mirror and disposed above polarizing half mirror **240**. For this reason, the video light that passes through polarizing half mirror **240** and travels upward is reflected by second reflecting mirror **260**. Second reflecting mirror **260** is disposed in an orientation in which second reflecting mirror **260** reflects such video light toward opening **221**. Stated differently, the video light reflected by second reflecting mirror **260** travels toward the eyes of the driver seated in the driver's seat via opening **221** to be second virtual image **201**. FIG. 1 shows the position of second virtual image **201** that is seen from the viewpoint of the driver. It is possible to adjust this position by adjusting the imaging position of the video light emitted from display element **230** of second display device **200**.

[Controller]

[0143] As shown in FIG. 1, controller **500** is electrically connected to first display device **100** and second display device **200** to collectively control the display contents of display elements **130** and **230**. More specifically, controller **500** includes a CPU, a RAM, a ROM, etc., and each process is executed by the CPU loading, in the RAM, a program stored in the ROM and executing such program.

[Positional Relationship Between First Virtual Image and Second Virtual Image]

[0144] The following describes the positional relationship between first virtual image **101** and second virtual image **201**. As shown in FIG. 1, first virtual image **101** is formed to be at an angle relative to the horizontal plane in a side view of vehicle **1**. More specifically, first virtual image **101** is located at an angle such that the lower end of first virtual image **101** is at the rearmost point in the traveling direction of vehicle **1** and the upper end of first virtual image **101** is at the foremost point in the traveling direction of vehicle **1**.

[0145] Second virtual image **201** is located below first virtual image **101** in a side view of vehicle **1**. Second virtual image **201** is formed to be parallel to the vertical plane of vehicle **1**. The lower end of first virtual image **101** and the upper end of second virtual image **201** are located within circle C whose diameter is the length of second virtual image **201** in a side view. The length of second virtual image **201** is the entire length of second virtual image **201** in a side view of vehicle **1**. In the present embodiment, since second virtual image **201** is located parallel to the vertical plane, the length of second virtual image **201** is the length in the vertical direction (up-down direction).

[0146] As described above, since the lower end of first virtual image **101** and the upper end of second virtual image **201** are located within circle C in a side view, it is possible to reduce each of the gap in the up-down direction and gap D in the horizontal direction between the lower end of first virtual image **101** and the upper end of second virtual image **201**. Here, the horizontal direction is the horizontal direction in a side view of vehicle **1**, that is, the front-back direction of vehicle **1**.

[0147] Here, when assuming that difference  $\Delta\theta$  between first angle of depression  $\theta_1$  formed by the lower end of first virtual image **101** with respect to the viewpoint of the driver and second angle of depression  $\theta_2$  formed by the upper end of second virtual image **201** with respect to the viewpoint of the driver is  $12^\circ$  or less, the lower end of first virtual image **101** and the upper end of second virtual image **201** are located within circle C in a side view and the gap in the up-down direction between the lower end of first virtual image **101** and the upper end of second virtual image **201** can be further reduced.

[0148] Note that, to further reduce the gap in the up-down direction between the lower end of first virtual image **101** and the upper end of second virtual image **201**, difference  $\Delta\theta$  may be, for example,  $10^\circ$  or less, and more preferably  $6^\circ$  or less.

Effects, Etc.

[0149] As described above, according to the present embodiment, the lower end of first virtual image **101** and the upper end of second virtual image **201** are located within circle C whose diameter is the length of second virtual image **201** in a side view. This enables the lower end of first virtual image **101** and the upper end of second virtual image **201** to be located in proximity to each other in the up-down direction. With this, each of the gap in the up-down direction and gap D in the horizontal direction between the lower end of first virtual image **101** and the upper end of second virtual image **201** is reduced. It is thus possible to alleviate the burden on the driver caused by moving his/her viewpoint and focus point, switching between first virtual image **101** and second virtual image **201**, in viewing these virtual images. Note that, from the standpoint of reducing the foregoing gaps, the diameter of circle C may be, for example, a half of the entire length of second virtual image **201**, and more preferably a quarter of the entire length.

[0150] Also, since difference  $\Delta\theta$  between first angle of depression  $\theta_1$  formed by the lower end of first virtual image **101** with respect to the viewpoint of the driver and second angle of depression  $\theta_2$  formed by the upper end of second virtual image **201** with respect to the viewpoint of the driver is  $12^\circ$  or less, it is possible to further reduce the gap in the up-down direction between the lower end of first virtual image **101** and the upper end of second virtual image **201**. This further alleviates the burden on the driver caused by moving his/her viewpoint and focus point, switching between first virtual image **101** and second virtual image **201**, in viewing these virtual images.

[0151] Also, since controller **500** controls the display content of first display device **100** and the display content of second display device **200**, it is possible for controller **500** to collectively control the display content of first display device **100** (display content of first virtual image **101**) and the display content of second display device **200** (display content of second virtual image **201**). This facilitates the content coordination between first virtual image **101** and second virtual image **201**.

Embodiment 2

[0152] The following describes Embodiment 2. In the following description, the same reference signs are assigned to the same parts as those of Embodiment 1, and the description of these parts may be omitted.

[0153] FIG. 4 is a schematic diagram showing display system **10A** according to Embodiment 2. As shown in FIG. 4, display system **10A** further includes third display device **300** that projects third virtual image **301** ahead of the driver. Third display device **300** is disposed in the vicinity of the upper end of windshield **2** and serves as an electronic inner mirror capable of displaying, as third virtual image **301**, camera video of the rear of the vehicle.

[0154] FIG. 5 is a schematic cross-sectional diagram showing third display device **300** according to Embodiment 2. As shown in FIG. 5, third display device **300** includes housing **320**, display element **330**, polarizing half mirror **340**, and reflecting mirror **350**.

[0155] Housing **320** is a box-shaped body formed of lightproof resin or metal. Opening **321** that faces rearward (defined as the rightward direction in FIG. 5) is provided in an upper end portion (defined as the upper side in FIG. 5) of the rear part (defined as the right side in FIG. 5) of housing **320**. Video light to serve as third virtual image **301** is projected from opening **321**. The internal space defined by housing **320** accommodates display element **330**, polarizing half mirror **340**, and reflecting mirror **350**.

[0156] Display element **330** is, for example, a liquid-crystal panel. When irradiated with light from a light source not shown, display element **330** emits, to polarizing half mirror **340**, video light to serve as third virtual image **301**. Display element **330** may also be an organic EL panel. Display element **330** is disposed in an orientation in which the display surface faces upward (defined as the upper side in FIG. 5). Although not shown, a  $\lambda/4$  plate is stacked on the display surface of display

element **330**.

[0157] Polarizing half mirror **340** is configured to reflect S-polarized light and transmit P-polarized light. More specifically, in polarizing half mirror **340**, a reflective polarizing plate is disposed on a glass substrate having a flat plate shape. Polarizing half mirror **340** is disposed in an orientation in which polarizing half mirror **340** reflects the video light outputted from display device **330** toward reflecting mirror **350**. Although not shown, a  $\lambda/4$  plate is stacked on the surface of polarizing half mirror **340** that faces display element **330**.

[0158] Reflecting mirror **350** is a concave mirror and is disposed in the forward direction of polarizing half mirror **340**. Reflecting mirror **350** is disposed in an orientation in which the concave surface serving as the reflecting surface faces rearward. Reflecting mirror **350** is disposed in an orientation in which reflecting mirror **350** reflects the video light that has transmitted through polarizing half mirror **340** toward opening **321**.

[0159] The P-polarized video light emitted from display element **330** enters the  $\lambda/4$  plate to be converted into circularly polarized light, and then travels toward polarizing half mirror **340**. Such video light enters the  $\lambda/4$  plate on polarizing half mirror **340** to be converted into S-polarized light, and then reflected by polarizing half mirror **340**. After that, the video light further enters the  $\lambda/4$  plate to be converted into circularly polarized light and then travels toward reflecting mirror **350** to be reflected by reflecting mirror **350**. Subsequently, the video light enters the  $\lambda/4$  plate on polarizing half mirror **340** to be converted into P-polarized light, and then transmits through polarizing half mirror **340** to travel toward opening **321**. The video light that has passed through opening **321** travels toward the eyes of the driver seated in the driver's seat to be third virtual image **301** (see FIG. 4).

[0160] Here, when third virtual image **301** is located at a viewing distance of 0.25 diopters or less with respect to the viewing distance of second virtual image **201**, it is possible to reduce the gap in the horizontal direction between second virtual image **201** and third virtual image **301**. It is thus possible to alleviate the burden on the driver caused by moving his/her viewpoint and focus point, switching between second virtual image **201** and third virtual image **301**, in viewing these virtual images. Note that “diopter” is defined as the absolute value of the difference between the reciprocal of the distance from the viewpoint to the left end of second virtual image **201** and the reciprocal of the distance from the viewpoint to the right end of third virtual image **301**.

[0161] FIG. 6 is a schematic diagram showing third display device **300a** according to Variation 1 of Embodiment 2. As shown in FIG. 6, third display device **300a** is disposed in the vicinity of the center of the lower end of windshield **2** and serves as a car navigation device that displays, for example, car navigation information as third virtual image **301a**.

[0162] FIG. 7 is a schematic diagram showing third display devices **300b** according to Variation 2 of Embodiment 2. As shown in FIG. 7, third display devices **300b** are disposed on the both sides of the lower end of windshield **2** and serve as electronic inner mirrors capable of displaying, as third virtual images **301b**, camera video of the side rear of vehicle **1**.

### Embodiment 3

[0163] The following describes Embodiment 3. FIG. 8 is a schematic diagram showing second display device **200B** according to Embodiment 3. As shown in FIG. 8, second display device **200B** includes three first reflecting mirrors **250b**. Three first reflecting mirrors **250b** are arranged along the longitudinal direction (normal direction on the plane of the diagram) of display element **230**. Of these three first reflecting mirrors **250b**, first reflecting mirror **250b** in the middle is located at the bottommost in FIG. 8, and the other two first reflecting mirrors **250b** are located above first reflecting mirror **250b** in the middle and at the same position in a side view in FIG. 8.

[0164] FIG. 9 is an explanatory diagram showing an example of the display performed by display element **230** according to Embodiment 3. As shown in FIG. 9, display element **230** is divided into three parts in the longitudinal direction, and these parts form different videos G1, G2, and G3. Here, video G2 at the center (which shows, for example, the state of the cruise control of the

vehicle) corresponds to first reflecting mirror **250b** in the middle, and the other two videos **G1** and **G3** (which show, for example, the speedometer and the engine tachometer) correspond to the other two first reflecting mirrors **250b**. In FIG. **8**, video light of video **G2** at the center is indicated by dashed lines, and each video light of the other two videos **G1** and **G3** is indicated by dash-dotted lines. Since first reflecting mirror **250b** in the middle is located in a different position from those of the other two first reflecting mirrors **250b**, the viewing distance of the second virtual image formed by video **G2** is different from the viewing distance of the second virtual images formed by videos **G1** and **G3**. More specifically, the viewing distance of the second virtual image formed by video **G2** is longer than the viewing distance of the second virtual images formed by videos **G1** and **G3**. Stated differently, in the eyes of the driver, the second virtual image formed by video **G2** appears to be present further than the second virtual images formed by videos **G1** and **G3**. As described above, since the second virtual images are displayed in a plurality of layers in the eyes of the driver, it is possible for second display device **200B** to provide diverse content displays.

[0165] Note that first virtual images may be displayed in a plurality of layers also in the first display device. FIG. **10** is a schematic diagram showing first display device **100B** according to Embodiment 3. As shown in FIG. **10**, first display device **100B** includes a plurality of (e.g., two) display elements **131b** and **132b**. Display element **131b** is located closer to first optical element **140** than display element **132b** is. For this reason, the viewing distance of the first virtual image formed by display element **131b** is shorter than the viewing distance of the first virtual image formed by display element **132b**. As a result, the first virtual image formed by display element **131b** and the first virtual image formed by display element **132b** are displayed in a plurality of layers in the eyes of the driver.

#### Embodiment 4

[0166] The following describes Embodiment 4. FIG. **11** is a schematic diagram showing second display device **200C** according to Embodiment 4. As shown in FIG. **11**, second reflecting mirror **260c** of second display device **200C** is a concave mirror. As described above, of the optical systems included in second display device **200C**, second reflecting mirror **260c** that reflects the video light at the final stage is a concave mirror. With this, the second virtual image is formed by means of the concave mirror reflecting and converging the video light at the final stage, thus easily securing the viewing distance of the second virtual image.

#### Embodiment 5

[0167] The following describes Embodiment 5. FIG. **12** is a schematic diagram showing first display device **100D** according to Embodiment 5. As shown in FIG. **12**, first display device **100D** includes adjuster **190** for adjusting the viewing distance of first virtual image **101**. Adjuster **190** is a device for moving display element **130** in the optical axis direction (normal direction of the display surface of display element **130**). Adjuster **190** includes, for example, a moving mechanism for moving display element **130** back and forth in the optical axis direction, on the basis of a driving source and power from such driving source. Stated differently, adjuster **190** adjusts the position of display element **130** with respect to the optical axis direction, thereby adjusting the viewing distance of first virtual image **101**.

[0168] FIG. **13** is a schematic diagram showing second display device **200D** according to Embodiment 5. As shown in FIG. **13**, second display device **200D** includes adjuster **290** for adjusting the viewing distance of second virtual image **201**. Adjuster **290** is a device for moving first reflecting mirror **250** in the optical axis direction of the video light to be reflected. Adjuster **290** includes, for example, a moving mechanism for moving first reflecting mirror **250** back and forth in the optical axis, on the basis of a driving source and power from such driving source. Stated differently, adjuster **290** adjusts the position of first reflecting mirror **250** with respect to the optical axis direction, thereby adjusting the viewing distance of second virtual image **201**.

[0169] As described above, since adjusters **190** and **290** are provided in first display device **100D** and second display device **200D**, respectively, it is possible to adjust the viewing distances of first



virtual image **101** and second virtual image **201**. With this, it is possible to adjust the viewing distances of first virtual image **101** and second virtual image **201** by means of adjusters **190** and **290**, in accordance with various conditions. This enables more diverse content representation. Note that it suffices if adjuster **190** and adjuster **290** are provided in at least one of first display device **100D** or second display device **200D**.

#### Embodiment 6

[0170] The following describes Embodiment 6. Embodiment 1 has described an example case where first virtual image **101** is located at an angle relative to the horizontal plane of vehicle **1**, and second virtual image **201** is located parallel to the up-down direction of vehicle **1**, as shown in FIG. **1**. Embodiment 6 describes other examples of the layout of the first virtual image and the second virtual image. FIGS. **14A** to **14C** are explanatory diagrams showing examples of the layout of the first virtual image and the second virtual image according to Embodiment 6.

[0171] In Example Layout **1** shown in FIG. **14A**, first virtual image **101e1** and second virtual image **201e1** are both located parallel to the up-down direction of vehicle **1** and are flush with each other. In this case, too, the lower end of first virtual image **101e1** and the upper end of second virtual image **201e1** are located within circle C.

[0172] In Example Layout **2** shown in FIG. **14B**, first virtual image **101e2** and second virtual image **201e2** are both located at an angle relative to the horizontal plane of vehicle **1** such that the upper ends are located above the lower ends. Furthermore, first virtual image **101e2** and second virtual image **201e2** are located flush with each other. In this case, too, the lower end of first virtual image **101e2** and the upper end of second virtual image **201e2** are located within circle C.

[0173] In Example Layout **3** shown in FIG. **14C**, first virtual image **101e3** is located at an angle relative to the horizontal plane of vehicle **1** such that first portion **101e31**, which is the lower end portion, is parallel to the up-down direction of vehicle **1**, and second portion **101e32**, which is located above such first portion **101e31**, is located upward, toward the front of vehicle **1**. Second virtual image **201e3** is located to be parallel to the vertical plane of vehicle **1** and flush with first portion **101e31** of first virtual image **101e3**. In this case, too, the lower end of first virtual image **101e3** and the upper end of second virtual image **201e3** are located within circle C.

[0174] FIG. **15** is a schematic diagram showing first display device **100E** corresponding to Example Layout **3** of Embodiment 6. As shown in FIG. **15**, first display device **100E** includes a plurality of (e.g., two) display elements **131e** and **132e**. Display elements **131e** and **132e** are disposed to be in mutually different orientations. Display element **131e** forms first portion **101e31** of first virtual image **101e3**, and display element **132e** forms second portion **101e32** of first virtual image **101e3**.

[0175] As described above, it is possible, in all of the example layouts, to reduce each of the gap in the up-down direction and the gap in the horizontal direction between the lower end of the first virtual image and the upper end of the second virtual image. This alleviates the burden on the driver caused by moving his/her viewpoint and focus point, switching between the first virtual image and the second virtual image, in viewing these virtual images.

#### Embodiment 7

[0176] The following describes Embodiment 7. Embodiment 1 has described an example case where each of first display device **100** and second display device **200** includes a reflective optical system. However, at least one of the first display device or the second display device may include an optical system that includes hologram elements. Embodiment 7 describes the case where the first display device includes an optical system which includes hologram elements.

[0177] FIG. **16** is a schematic diagram showing first display device **100F** according to Embodiment 7. As shown in FIG. **16**, first display device **100F** includes image generating device **155f** and light guide body **160f**.

[0178] Image generating device **155f** is a device that outputs video light to light guide body **160f**. Light guide body **160f** is an optical system that includes hologram elements, and is a hologram

light guide body in the present embodiment. Light guide body **160f**, which has light-transmissivity, enlarges the video light outputted by image generating device **155f** and outputs the enlarged video light toward windshield **2**. Light guide body **160f** includes light guide **161f** having light-transmissivity, and a plurality of hologram elements **162f**.

[0179] Light guide **161f** includes input surface **163f** facing image generating device **155f** and output surface **164f** facing windshield **2**. Light guide **161f** is configured using a material having light-transmissivity such as glass and a resin material.

[0180] Light guide **161f** internally includes the plurality of hologram elements **162f**. The plurality of hologram elements **162f** are light-transmissive optical elements that diffract light propagating inside light guide **161f** and outputs the diffracted light. The plurality of hologram elements **162f** are internally included in light guide **161f** in an orientation that is approximately parallel to the input surface and the output surface of light guide **161f**. The plurality of hologram elements **162f** are configured using a material having light-transmissivity. The plurality of hologram elements **162f** include first hologram element **165f**, second hologram element **166f**, and third hologram element **167f**.

[0181] First hologram element **165f** is an input hologram element where the video light outputted by image generating device **155f** enters. First hologram element **165f** outputs the video light that has entered first hologram element **165f** toward second hologram element **166f**. More specifically, first hologram element **165f** deflects the video light by diffraction in accordance with the diffraction efficiency of first hologram element **165f** and outputs the deflected video light toward second hologram element **166f**.

[0182] Second hologram element **166f** is a folding hologram element that diffracts the video light from first hologram element **165f** and outputs the diffracted video light toward third hologram element **167f**. More specifically, second hologram element **166f** deflects the video light by diffraction in accordance with the diffraction efficiency of second hologram element **166f** and outputs the deflected video light toward third hologram element **167f**.

[0183] Third hologram element **167f** diffracts the video light from second hologram element **166f** and outputs, from output surface **164f**, the diffracted video light to outside of light guide body **160f**. More specifically, third hologram element **167f** deflects the video light by diffraction in accordance with the diffraction efficiency of third hologram element **167f** and outputs the deflected video light to outside from output surface **164f**.

[0184] As described above, since first display device **100F** includes an optical system (light guide body **160f**) that includes the hologram elements, it is possible for first display device **100F** to have a low-profile optical system.

#### Embodiment 8

[0185] The following describes Embodiment 8. Embodiment 1 has shown an example case where a first display position of first virtual image **101** and a second display position of second virtual image **201** are not changeable. However, the first display position of the first virtual image and the second display position of the second virtual image may be changeable.

[0186] FIG. **17** is a schematic diagram showing display system **10G** according to Embodiment 8. FIG. **18** is an explanatory diagram showing first virtual image **101g** and second virtual image **201g** in the reference positions in display system **10G** according to Embodiment 8. Note that, in a strict sense, FIG. **18** shows displayable range **R1g** of first virtual image **101g** and displayable range **R2g** of second virtual image **201g**. The displayable ranges are regions where virtual images are displayable.

[0187] As shown in FIG. **17**, display system **10G** includes first position adjuster **171g** and second position adjuster **172g**, which are electrically connected to and controlled by controller **500g**.

[0188] First position adjuster **171g** is provided in housing **110** of first display device **100G** and adjusts the first display position in the up-down direction of first virtual image **101g**. More specifically, first position adjuster **171g** includes a rotation mechanism for adjusting the orientation

(tilt) of second optical element **150g** and a drive motor for driving such rotation mechanism. Second optical element **150g** is an example of the first reflector that reflects the video light forming first virtual image **101g**. Second optical element **150g** rotates clockwise or counterclockwise, with the center of its reflecting surface, for example, serving as the center of rotation. This movement is caused by the rotation mechanism and the drive motor included in first position adjuster **171g**. This causes the orientation of second optical element **150g** to change. In response to the change in the orientation of second optical element **150g**, the optical path of the video light forming first virtual image **101g** also moves. As a result, the first display position of first virtual image **101g** is adjusted. [0189] Second position adjuster **172g** is provided in housing **220** of second display device **200G** and adjusts the second display position in the up-down direction of second virtual image **102g**. More specifically, second position adjuster **172g** includes a rotation mechanism for adjusting the orientation (tilt) of second reflecting mirror **260g** and a drive motor for driving such rotation mechanism. Second reflecting mirror **260g** is an example of the second reflector that reflects the video light forming second virtual image **201g**. Second reflecting mirror **260g** rotates clockwise or counterclockwise, with the center of its reflecting surface, for example, serving as the center of rotation. This movement is caused by the rotation mechanism and the drive motor included in second position adjuster **172g**. This causes the orientation of second reflecting mirror **260g** to change. In response to the change in the orientation of second reflecting mirror **260g**, the optical path of the video light forming second virtual image **201g** also moves. As a result, the second display position of second virtual image **201g** is adjusted.

[0190] Further, as shown in FIG. **18**, display system **10G** includes head imager **175g** for imaging the head of the driver. Head imager **175g** is a camera disposed in the vicinity of the upper portion of windshield **2** in vehicle **1**, and captures an image of the vehicle interior. Head imager **175g** is connected to controller **500g**, and outputs the captured image to controller **500g**. Controller **500g** detects the head position of the driver from the image captured by head imager **175g**, and estimates the viewpoint position of the driver, on the basis of such detected head position. Stated differently, controller **500g** is an example of the estimator that estimates the viewpoint position of the driver, on the basis of the image captured by head imager **175g**.

[0191] Controller **500g** controls first position adjuster **171g** and second position adjuster **172g**, on the basis of the estimated viewpoint position of the driver. As is known from the above, controller **500g** is an example of the adjustment controller. Through this control, it is possible to adjust the first display position and the second display position to positions that are responsive to the viewpoint position of the driver.

[0192] Subsequently, controller **500g** changes the coordinates of first virtual image **101g** within displayable range **R1g** of first virtual image **101g**, on the basis of the first display position and the second display position that have been adjusted. With this, it possible to display first virtual image **101g** and second virtual image **201g** for the driver to have a lesser sense of discomfort.

[0193] The following specifically describes the difference between first virtual image **101g** and second virtual image **201g** before and after adjusting the first display position and the second display position.

[0194] FIG. **19** is an explanatory diagram showing first virtual image **101g** and second virtual image **201g** in the reference positions seen from the driver in Embodiment 8. As shown in FIG. **19**, displayable range **R1g** of first virtual image **101g** is indicated by single dash-dotted lines, and displayable range **R2g** of second virtual image **201g** is indicated by double dash-dotted lines. First virtual image **101g** is represented as the dot-hatched region within displayable range **R1g**. Second virtual image **201g** is represented as the hatched region within displayable range **R2g**. Second virtual image **201g** is located across the entirety of displayable range **R2g**. The lower end portion of displayable range **R1g** overlaps displayable range **R2g**, and first virtual image **101g** is not displayed in such overlapping portion. Stated differently, first virtual image **101g** and second virtual image **201g** are displayed without any gaps in the up-down direction. With this, first virtual image **101g**

and second virtual image **201g** appear to be integrated in the eyes of the driver, thus enabling the driver to be less likely to feel a sense of discomfort.

[0195] Next, the case is assumed where a large driver takes over driving. FIG. **20** is an explanatory diagram showing a state in which a large driver takes over driving, and the first display position and the second display position have been adjusted in Embodiment 8. FIG. **20** corresponds to FIG. **18**.

[0196] As shown in FIG. **20**, since a large driver has taken over driving, the viewpoint position also moves upward (from “viewpoint before change of driver” to “viewpoint after change of driver” in FIG. **20**). Controller **500g** detects the head position of the driver, on the basis of the image captured by head imager **175g**, and estimates the viewpoint position of the driver, on the basis of such detected head position. After that, controller **500g** controls first position adjuster **171g** and second position adjuster **172g**, on the basis of the estimated viewpoint position of the driver. Through this, the first display position and the second display position are adjusted to positions that are responsive to the viewpoint position of the driver. In the present embodiment, the first display position is moved upward without moving the second display position. As a result, displayable range **R1g** is moved up to a position at which displayable range **R1g** does not overlap displayable range **R2g**. At this time, the smaller the better the gap between the lower end of displayable range **R1g** and the upper end of displayable range **R2g**.

[0197] FIG. **21A** is an explanatory diagram showing first virtual image **101g** and second virtual image **201g** only after the first display position has been adjusted in Embodiment 8. FIG. **21A** corresponds to FIG. **19**. As shown in FIG. **21A**, the mere movement of the first display position upward results in the coordinates of first virtual image **101g** within displayable range **R1g** remaining constant. As a result, a gap occurs between first virtual image **101g** and second virtual image **201g**.

[0198] For this reason, controller **500g** changes the coordinates of first virtual image **101g** within displayable range **R1g** of first virtual image **101g**, on the basis of the first display position and the second display position that have been adjusted. More specifically, controller **500g** causes first virtual image **101g** to be displayed at coordinates, within displayable range **R1g**, that are closer to displayable range **R2g** than before the coordinates are changed. More preferably, controller **500g** causes first virtual image **101g** to be displayed at coordinates at which the lower end of first virtual image **101g** overlaps the lower end of displayable range **R1g**.

[0199] FIG. **21B** is an explanatory diagram showing first virtual image **101g** and second virtual image **201g** in the case where the first display position has been adjusted and the coordinates of first virtual image **101g** have been adjusted in Embodiment 8. FIG. **21B** corresponds to FIG. **21A**. As shown in FIG. **21B**, since the coordinates of first virtual image **101g** within displayable range **R1g** have been adjusted as described above, the gap between first virtual image **101g** and second virtual image **201g** has been reduced. With this, first virtual image **101g** and second virtual image **201g** appear to be integrated in the eyes of the driver, thus enabling the driver to be less likely to feel a sense of discomfort.

[0200] As described above, display system **10G** includes controller **500g**, which is the coordinate changer, and first position adjuster **171g** and second position adjuster **172g**. With this, even when the viewpoint position of the driver has changed, it is possible to adjust the coordinates of first virtual image **101g** within displayable range **R1g**, while adjusting the first display position of first virtual image **101g** and the second display position of second virtual image **201g**. This enables first virtual image **101g** and second virtual image **201g** to be suitably displayed also for a different driver.

[0201] Further, since first position adjuster **171g** adjusts the first display position of first virtual image **101** by adjusting the orientation (tilt) of second optical element **150g**, it is possible to adjust the first display position using a simple structure. Meanwhile, since second position adjuster **172g** adjusts the second display position of second virtual image **201g** by adjusting the orientation (tilt)

of second reflecting mirror **260g**, it is possible to adjust the second display position using a simple structure. These enable first virtual image **101g** and second virtual image **201g** to be suitably displayed also for a different driver.

[0202] Further, controller **500g** estimates the viewpoint position of the driver, on the basis of the image captured by head imager **175g**, and controls first position adjuster **171g** and second position adjuster **172g**, on the basis of such estimated viewpoint position. This enables the first display position and the second display position to be automatically adjusted.

[0203] Note that the present embodiment has described an example case where first position adjuster **171g** adjusts the first display position by adjusting the orientation of second optical element **150g**. However, first position adjuster **171g** may adjust the first display position by adjusting the position of second optical element **150g**. Further, when the first display position is adjustable, first position adjuster **171g** may adjust at least one of the orientation or the position of another optical element (display element **130**, first optical element **140**, etc.), or may adjust the orientation and the position of the entirety of first display device **100G**. The same applies to second position adjuster **172g**.

[0204] The present embodiment has also described an example case where controller **500g** automatically adjusts the first display position and second display position, but these adjustments may also be manually made. More specifically, display system **10G** may include a control unit for inputting operation instructions to be provided to first position adjuster **171g** and second position adjuster **172g**. When the user operates the control unit to adjust the first display position and the second display position, for example, controller **500g** controls first display device **100G** and second display device **200G** to cause them to display first virtual image **101g** and second virtual image **201g** for testing. Furthermore, when the user inputs an operation instruction to the control unit for moving the first display position and second display position, controller **500g** causes first position adjuster **171g** and second position adjuster **172g** to operate, on the basis of such operation instruction. With this, it is possible to adjust the first display position and the second display position to the positions intended by the user. In this case, too, controller **500g** changes the coordinates of first virtual image **101g** within displayable range **R1g** of first virtual image **101g**, on the basis of the first display position and the second display position that have been adjusted.

#### Embodiment 9

[0205] FIG. **22** is a schematic diagram showing display system **A10** according to Embodiment 9 disposed in vehicle **A1**. In FIG. **22**, vehicle **A1** is shown in cross-section. Here, the user of display system **A10** is the driver in vehicle **A1**.

[0206] As shown in FIG. **22**, display system **A10** includes first display device **A100**, second display device **A200**, first imager **A400**, and controller **A500**. First display device **A100** and second display device **A200** are disposed, for example, inside dashboard **A4** of vehicle **A1** (see FIG. **23**). First display device **A100** and second display device **A200** display, for example, vehicle information related to vehicle **A1** as first virtual image **A101** and second virtual image **A201**. Examples of the vehicle information include the vehicle speed of vehicle **A1**, the number of revolutions of the engine, a detection result on an object that is present in the proximity of vehicle **A1**, navigation information from the current position of vehicle **A1** to the destination, image information captured by a camera that captures an image of the rear of vehicle **A1**, notification information to be provided to the driver, etc. The notification information is information indicating that target object **AP** (see FIG. **23**) is present in the image captured by first imager **A400**. Target object **AP** is, for example, a moving object (pedestrian, animal, vehicle other than vehicle **A1** such as automobile, motorcycle, kickboard, etc.) that is present in the surroundings of vehicle **A1**. Stated differently, the notification information is information for notifying the driver of vehicle **A1** of that a moving object is present in the proximity of vehicle **A1**.

[0207] Note that FIG. **22** shows an example case where first display device **A100** and second display device **A200** are disposed inside dashboard **A4**, but the position where first display device

**A100** and second display device **A200** are disposed is not limited to this; first display device **A100** may be disposed, for example, in the vicinity of the upper end of windshield **A2** and second display device **A200** may be disposed, for example, in the center console.

[0208] First display device **A100** is a transmissive display such as an augmented reality head-up display (AR-HUD). First display device **A100** projects light onto windshield (windscreen) **A2** serving as a display medium. The projected light is reflected by windshield **A2**. Such reflected light travels toward the eyes of the driver, who is seated in the driver's seat. The driver perceives the reflected light that has entered his/her eyes as first virtual image **A101** that is seen on the opposite side across windshield **A2** (outside the vehicle), against the actual objects seen across windshield **A2** serving the background. FIG. 22 shows an example of the position of first virtual image **A101** seen from the viewpoint of the driver. It is possible to set this position by adjusting the imaging position of the video light outputted from first display device **A100**. The viewpoint of the driver is, for example, the reference eyepoint. The reference eyepoint is "the point that represents the position of the eyes of the driver under normal driving conditions".

[0209] Second display device **A200** is a non-transmissive display. Second display device **A200** projects the video light toward the driver. The driver perceives the video light that has entered his/her eyes as second virtual image **A201** that appears in a distant position from second display device **A200**. Stated differently, the video light outputted from second display device **A200** travels toward the eyes of the driver seated in the driver's seat to be second virtual image **A201**. FIG. 22 shows an example of the position of second virtual image **A201** seen from the viewpoint of the driver. Note that second display device **A200** may be a transmissive display such as a head-up display (HUD).

[0210] The following describes the positional relationship between first virtual image **A101** and second virtual image **A201**. As shown in FIG. 22, first virtual image **A101** is formed to be parallel to the vertical plane in a side view of vehicle **A1**.

[0211] Second virtual image **A201** is located below first virtual image **A101** in a side view of vehicle **A1**. Second virtual image **A201** is formed to be parallel to the vertical plane. The lower end of first virtual image **A101** and the upper end of second virtual image **A201** are located in proximity to each other. More specifically, the lower end of first virtual image **A101** and the upper end of second virtual image **A201** are located within circle **AC** whose diameter is the length of second virtual image **A201** in a side view. The length of second virtual image **A201** is the entire length of second virtual image **A201** in a side view. In the present embodiment, since second virtual image **A201** is located parallel to the vertical plane, the length of second virtual image **A201** is the length in the vertical direction (up-down direction).

[0212] As described above, since the lower end of first virtual image **A101** and the upper end of second virtual image **A201** are located in proximity to each other, it is possible to reduce each of the gap in the up-down direction and the gap in the horizontal direction between the lower end of first virtual image **A101** and the upper end of second virtual image **A201**.

[0213] First imager **A400** is a camera that captures images of the surroundings of vehicle **A1**. More specifically, first imager **A400** captures images of an area ahead of vehicle **A1**. First imager **A400** is disposed more forward than the driver is who is seated in the driver's seat. More specifically, first imager **A400** is disposed at the front-end portion of vehicle **A1**. This enables first imager **A400** to image target object **AP** that is in a blind spot of the driver.

[0214] Controller **A500** is disposed inside dashboard **A4** and controls first display device **A100**, second display device **A200**, and first imager **A400**. Since a single controller **A500** controls first display device **A100**, second display device **A200**, and first imager **A400**, it is possible to simplify the control structure of the entire system and also reduce power consumption.

[0215] More specifically, controller **A500** includes a CPU, a RAM, a ROM, etc., and each process is executed by the CPU loading, in the RAM, a program stored in the ROM and executing such program. For example, controller **A500** performs image processing on image data captured by first

imager **A400** to recognize at least one target object **AP** included in such image data, and detects the position coordinates, the size, etc. of such target object **AP**. Controller **A500** further performs image processing on the image data to recognize a structure (building, wall, etc.) included in such image data, and detects the position coordinates, the size, etc. of the structure. Controller **A500** extracts target object **AP** that is in a blind spot due to the structure when seen from the viewpoint of the driver (reference eyepoint), and generates notification information **AY110** and **AY210** (see FIG. 23), on the basis of, for example, the position coordinates and the size of such extracted target object **AP**. Controller **A500** then causes first display device **A100** and second display device **A200** to display the notification information.

#### Example Display

[0216] The following describes an example of displaying notification information **AY110** and **AY210**. FIG. 23 is an explanatory diagram showing an example of displaying notification information **AY110** and **AY210** according to Embodiment 9. In the example display shown in FIG. 23, the upper half of first display area **A102** of first virtual image **A101** displayed by first display device **A100** mostly overlaps windshield **A2**, and the remaining part overlaps hood **A3** of vehicle **A1**. Second display area **A202** of second virtual image **A201** displayed by second display device **A200** overlaps dashboard **A4**.

[0217] Controller **A500** controls first display device **A100** to cause it to display notification information **AY110** in the form of an arrow as first virtual image **A101** within first display area **A102**. Meanwhile, controller **A500** controls second display device **A200** to cause it to display the video captured by first imager **A400** as second virtual image **A201** across the entirety of second display area **A202**, and also to display notification information **AY210** in the form of an arrow as second virtual image **A201** within second display area **A202**. Stated differently, in second display area **A202**, notification information **AY210** is displayed on the video captured by first imager **A400** in an overlaid manner. Here, the video captured by first imager **A400** is an example of the vehicle surrounding information. The vehicle surrounding information is information indicating the situation around vehicle **A1**. In the present embodiment, the vehicle surrounding information is video, captured by first imager **A400**, of an area ahead of vehicle **A1**. Note that the vehicle surrounding information may be an image for navigation that is synchronized with the video captured by first imager **A400**, or may be a combination of the video and the image for navigation. When causing first display device **A100** and second display device **A200** to display notification information **AY110** and **AY210**, controller **A500** causes first display device **A100** and second display device **A200** to display notification information **AY110** and **AY210** after adjusting at least one of the luminance, the size, or the chromaticity of each of notification information **AY110** and **AY210**, and to display notification information **AY110** and **AY210** together with the vehicle surrounding information obtained from first imager **A400**. Details of these adjustments are described later.

[0218] Notification information **AY110** and **AY210** are graphics for guiding the line of sight of the driver to target object **AP**. The present embodiment shows an example case where notification information **AY110** and **AY210** are arrows, but notification information **AY110** and **AY210** may also be other graphics. To increase the effect of guiding the line of sight of the driver, it suffices if notification information **AY110** and **AY210** are displayed at least for a certain length of time (e.g., 0.2 seconds or longer). Notification information **AY110** and **AY210** may alternately flash. In the case where notification information **AY110** and **AY210** flash, the flashing cycle is, for example, between 0.5 seconds and 1 second, inclusive. The flashing cycle may be different for alerting and for warning. For example, the flashing cycle for warning that is shorter than the flashing cycle for alerting can express a sense of caution. In addition, one of notification information **AY110** and **AY210** may be kept lit while the other is flashing.

[0219] Target object **AP** is in a blind spot due to wall **AW** and thus invisible from the viewpoint of the driver. However, since first imager **A400** is capturing an image of such target object **AP**, target

object AP appears in the video captured by first imager A400. Controller A500 extracts target object AP that is in the blind spot and causes notification information AY110 and AY210 to be displayed on straight line AL that connects the coordinate position of target object AP and display reference point APS. Straight line AL is a virtual straight line that is set spanning first display area A102 and second display area A202. Although not displayed in first display area A102 and second display area A202 in FIG. 23, straight line AL may be displayed.

[0220] Since the present embodiment shows the case where target object AP is present outside first display area A102, display reference point APS is set on center line ALc in the width direction of first display area A102 (width direction of the vehicle). Display reference point APS may be located either inside or outside first display area A102. Furthermore, display reference point APS may be located at the vanishing point seen from the driver. In this case, the driver is more likely to notice notification information AY110 when looking ahead.

[Luminance Adjustment]

[0221] When causing first display device A100 and second display device A200 to display notification information AY110 and AY210, controller A500 may adjust the luminance of each of notification information AY110 and AY210 between first display device A100 and second display device A200, on the basis of the vehicle surrounding information.

[0222] More specifically, controller A500 calculates the luminance of the background from the video, captured by first imager A400, of an area ahead of vehicle A1 (vehicle surrounding information). Controller A500 calculates the luminance of first virtual image A101 in a state of being overlaid on the background, on the basis of the luminance of such background. Stated differently, when the luminance of the background is high, such as during the daytime, controller A500 calculates the luminance of first virtual image A101, especially the luminance of notification information AY110 in first virtual image A101, such that the luminance is at or higher than the level at which the driver can view notification information AY110. Conversely, when the luminance of the background is low, such as at night time, controller A500 calculates the luminance of first virtual image A101, especially the luminance of notification information AY110 in first virtual image A101, such that the luminance is low to a degree at which the driver will not suffer glare. Controller A500 adjusts the luminance of each of notification information AY110 and AY210 between first display device A100 and second display device A200, on the basis of the calculation result.

[0223] For example, controller A500 may adjust the luminance of each of notification information AY110 and AY210 between first display device A100 and second display device A200 to cause the luminance of notification information AY110 in first virtual image A101 and the luminance of notification information AY210 in second virtual image A201 to be approximately the same. With this, it is possible to reduce the difference in the luminance between notification information AY110 in first virtual image A101 and notification information AY210 in second virtual image A201. This further reduces the user's sense of discomfort that occurs when the user views notification information AY110 in first virtual image A101 and notification information AY210 in second virtual image A201. It suffices if the difference in the luminance of notification information AY110 from the luminance of notification information AY210 is 20% or less, and more preferably 10% or less.

[0224] Controller A500 may also adjust the luminance of each of notification information AY110 and AY210 between first display device A100 and second display device A200 to cause the luminance of notification information AY110 in first virtual image A101 to be higher than the luminance of notification information AY210 in second virtual image A201. With this, since the luminance of notification information AY110 in first virtual image A101 is adjusted to be higher than the luminance of notification information AY210 in second virtual image A201, it is possible to make notification information AY110 in first virtual image A101, which is located above second virtual image A201, more stand out. This enables the driver to more easily recognize notification



information AY110 in first virtual image A101. The difference in the luminance of notification information AY110 from the luminance of notification information AY210 is preferably 20% or higher.

[0225] Controller A500 may also adjust the luminance of each of notification information AY110 and AY210 between first display device A100 and second display device A200 to cause the luminance of notification information AY110 in first virtual image A101 to be lower than the luminance of notification information AY210 in second virtual image A201. With this, since the luminance of notification information AY110 in first virtual image A101 is adjusted to be lower than the luminance of notification information AY210 in second virtual image A201, it is possible to make notification information AY210 in second virtual image A201, which is displayed below first virtual image A101 and thus directs the line of sight of the driver downward, more stand out. The difference in the luminance of notification information AY110 from the luminance of notification information AY210 is preferably 20% or higher.

[0226] Note that when the upper limit of the luminance of first display device A100 is preliminarily set by the user, it is preferable that controller A500 adjusts the luminance of notification information AY110 such that such luminance of first display device A100 does not exceed the upper limit of the luminance.

[Size Adjustment]

[0227] Controller A500 may adjust the size of each of notification information AY110 and AY210 between first display device A100 and second display device A200. More specifically, controller A500 adjusts the width of each of notification information AY110 and AY210 between first display device A100 and second display device A200 to cause the width of the arrow representing notification information AY110 and the width of the arrow representing notification information AY210 shown in FIG. 23 to be approximately the same. The difference in the width of notification information AY110 from the width of notification information AY210 is preferably is 10% or less.

[Chromaticity Adjustment]

[0228] When causing first display device A100 and second display device A200 to display notification information AY110 and AY210, controller A500 may adjust the chromaticity of each of notification information AY110 and AY210 between first display device A100 and second display device A200, on the basis of the vehicle surrounding information.

[0229] More specifically, controller A500 calculates the chromaticity of the background from the video, captured by first imager A400, of an area ahead of vehicle A1 (vehicle surrounding information). Controller A500 calculates the chromaticity of first virtual image A101 in a state of being overlaid on the background, on the basis of the chromaticity of such background. Controller A500 adjusts the chromaticity between first display device A100 and second display device A200 to cause the chromaticity of notification information AY110 in first virtual image A101 to be approximately the same as the chromaticity of notification information AY210 in second virtual image A201, on the basis of the calculation result. When the color of the background is mainly green, as in the case where the vehicle is traveling through a mountainous area, for example, notification information AY110 colored green will not stand out. For this reason, it suffices if the chromaticity of notification information AY110 is adjusted such that notification information AY110 appears red, which is the complementary color of green. To this end, by coloring notification information AY110 purple, it will be mixed with the green of the background, as a result of which notification information AY110 will appear red in the eyes of the driver, making it easier for the driver to recognize. When this is done, notification information AY210 of second display device A200 is simply required to be colored red. Here, the difference (color difference  $\Delta E^*$ ) between the chromaticity of notification information AY110 and the chromaticity of notification information AY210 is preferably 6.5 or less, and more preferably 3.2 or less.

[0230] In addition, controller A500 may adjust the chromaticity of notification information AY110 in first virtual image A101, with the chromaticity of notification information AY210 in second

virtual image **A201** kept constant. This enables notification information **AY110** to be more easily viewed in first virtual image **A101**.

[0231] In addition, controller **A500** may adjust the chromaticity of notification information **AY210** in second virtual image **A201** to be approximately the same as the chromaticity of notification information **AY110** in first virtual image **A101**, with the chromaticity of notification information **AY110** in first virtual image **A101** kept constant. This enables the chromaticity of notification information **AY110** in first virtual image **A101** to be less likely to change, thus alleviating the burden on the driver when looking ahead.

Effects, Etc.

[0232] As described above, according to the present embodiment, when notification information **AY110** and **AY210** are displayed by first display device **A100** and second display device **A200**, at least one of the luminance, the size, or the chromaticity of each of notification information **AY110** and **AY210** to be displayed together with the vehicle surrounding information obtained from first imager **A400** is adjusted between first display device **A100** and second display device **A200**. This enables notification information **AY110** displayed by first display device **A100** and notification information **AY210** displayed by second display device **A200** to be displayed in a visually uniform (consistent) manner. It is thus possible to reduce the driver's sense of discomfort that occurs when such driver views notification information **AY110** in first virtual image **A101** and notification information **AY210** in second virtual image **A201**.

[0233] First display device **A100** projects first virtual image **A101** and second display device **A200** projects second virtual image **A201**. As such, when the driver views notification information **AY110** in first virtual image **A101** and notification information **AY210** in second virtual image **A201**, these virtual images are viewed at distant positions, thus reducing the driver's sense of discomfort.

[0234] The lower end of first virtual image **A101** and the upper end of second virtual image **A201** are located in proximity to each other. It is thus possible to reduce each of the gap in the up-down direction and the gap in the horizontal direction between the lower end of first virtual image **A101** and the upper end of second virtual image **A201**. This alleviates the burden on the driver caused by moving his/her viewpoint and focus point when shifting the eyes from first virtual image **A101** to second virtual image **A201**, or from second virtual image **A201** to first virtual image **A101**.

[0235] The luminance of first virtual image **A101** in a state of being overlaid on the background is calculated, on the basis of the vehicle surrounding information. Then, on the basis of the calculation result, the luminance of each of notification information **AY110** and **AY210** is adjusted between first display device **A100** and second display device **A200**. With this, it is possible to reflect, on the luminance of each of notification information **AY110** and **AY210**, the influence of the background exerted on the luminance of first virtual image **A101**. This enables information notification that creates a lesser sense of discomfort.

[0236] The luminance of notification information **AY110** in first virtual image **A101** and the luminance of notification information **AY210** in second virtual image **A201** are adjusted to be approximately the same. It is thus possible to reduce the difference in the luminance between notification information **AY110** in first virtual image **A101** and notification information **AY210** in second virtual image **A201**. This further reduces the driver's sense of discomfort that occurs when such driver views notification information **AY110** in first virtual image **A101** and notification information **AY210** in second virtual image **A201**.

[0237] The luminance of notification information **AY110** in first virtual image **A101** is adjusted to be higher than the luminance of notification information **AY210** in second virtual image **A201**. It is thus possible to make notification information **AY110** in first virtual image **A101**, which is located above second virtual image **A201**, more stand out. This enables the driver to more easily recognize notification information **AY110** within first virtual image **A101**.

[0238] The luminance of notification information **AY110** in first virtual image **A101** is adjusted to

be lower than the luminance of notification information AY210 in second virtual image A201. It is thus possible to make notification information AY210 in second virtual image A201, which is displayed below first virtual image A101 and thus directs the line of sight of the driver downward, more stand out.

[0239] The size of each of notification information AY110 and AY210 is adjusted between first display device A100 and second display device A200. It is thus possible to reduce a sense of discomfort attributable to the size of each of notification information AY110 and AY210 between first display device A100 and second display device A200.

[0240] The chromaticity of first virtual image A101 in a state of being overlaid on the background is calculated, on the basis of the vehicle surrounding information. Then, on the basis of the calculation result, the chromaticity of notification information AY110 in first virtual image A101 and the chromaticity of notification information AY210 in second virtual image A201 are adjusted to be approximately the same. With this, it is possible to adjust the chromaticity of notification information AY110 in first virtual image A101 and the chromaticity of notification information AY210 in second virtual image A201 to be approximately the same, in consideration of the influence of the background exerted on first virtual image A101. This enables information notification that creates a lesser sense of discomfort.

[0241] The chromaticity of notification information AY110 in first virtual image A101 is adjusted, with the chromaticity of notification information AY210 in second virtual image A201 kept constant. It enables notification information AY110 to be more easily viewed in first virtual image A101.

[0242] The chromaticity of notification information AY210 in second virtual image A201 is adjusted to be approximately the same as the chromaticity of notification information AY110 in first virtual image A101, with the chromaticity of notification information AY110 in first virtual image A101 kept constant. It is thus possible for the chromaticity of notification information AY110 in first virtual image A101 to be less likely to change. This alleviates the burden on the driver when looking ahead.

[0243] When target object AP is present outside first display area A102, notification information AY110 is displayed on display reference point APS, which is set on center line ALc in the width direction of first display area A102. Stated differently, since notification information AY110 is located on center line ALc of first display area A102, when target object AP is not present in first display area A102, it is possible for the driver to more reliably view notification information AY110.

[0244] Notification information AY110 displayed by first display device A100 and notification information AY210 displayed by second display device A200 are located on straight line AL that includes display reference point APS. It is thus possible for the driver to linearly move the line of sight to each of notification information AY110 and AY210. This alleviates the burden on the driver caused by moving the line of sight.

[0245] The video captured by first imager A400 is displayed by second display device A200. It is thus possible to display the video captured by first imager A400 and notification information AY210 displayed by second display device A200 in an overlaid manner. This enables notification information AY210 to be displayed on a clear video, thereby enabling content representation that is easy for the driver to understand.

#### Embodiment 10

[0246] The following describes Embodiment 10. FIG. 24 is a schematic diagram showing display system A10A according to Embodiment 10. As shown in FIG. 24, display system A10A further includes first imager A400a that captures images of an area ahead of vehicle A1. First imager A400a is disposed in the vicinity of the upper end of windshield A2 inside vehicle A1, and captures images of the area ahead of vehicle A1 via windshield A2. Controller A500 calculates the luminance and the chromaticity of the background from the video, captured by first imager A400a,

of an area ahead of vehicle **A1** (vehicle surrounding information). Stated differently, in Embodiment 10, first imager **A400** is an imager dedicated to imaging an area ahead of vehicle **A1** and displaying the captured video on second display device **A200**, and first imager **A400a** is an imager dedicated to calculating the luminance and the chromaticity of the background. As described above, since first imager **A400a** dedicated to the calculation of the luminance and the chromaticity of the background is provided, it is possible to further improve the accuracy of luminance adjustment and chromaticity adjustment. In particular, since first imager **A400a** captures images of the area ahead of vehicle **A1** via windshield **A2** from inside the vehicle, it is possible to reflect, on the luminance adjustment and chromaticity adjustment, the luminance and the chromaticity of the background seen from inside the vehicle, which is preferable.

#### Embodiment 11

[0247] The following describes Embodiment 11. FIG. 25 is a schematic diagram showing display system **A10B** according to Embodiment 11. As shown in FIG. 25, display system **A10B** further includes second imager **A600b** that captures an image of the surroundings of the driver. Second imager **A600b** is disposed in the vicinity of the upper end of windshield **A2** and captures an image of an area in the vicinity of the headrest of the driver's seat. Stated differently, when the driver is seated in the driver's seat, second imager **A600b** captures an image of the head of the driver. Controller **A500** performs image processing, on the basis of the image captured by second imager **A600b**, to obtain information on the surroundings of the driver (user surrounding information). More specifically, controller **A500** performs image processing on the image captured by second imager **A600b** to detect the viewpoint position of the driver and obtains such viewpoint position as the user surrounding information. Controller **A500** estimates the luminance of the background when the driver is viewing first virtual image **A101**, on the basis of the viewpoint position. Controller **A500** reflects such estimation result on the luminance adjustment performed on notification information **AY110** and **AY210**. This achieves information notification that creates a lesser sense of discomfort.

[0248] In addition, controller **A500** may adjust the size of each of notification information **AY110** and **AY210** between first display device **A100** and second display device **A200**, on the basis of the user surrounding information obtained from second imager **A600b**. Controller **A500** estimates a first distance between the driver and first virtual image **A101** and a second distance between the driver and second virtual image **A201**, on the basis of the viewpoint position. Controller **A500** reflects the first distance and the second distance on the size adjustment performed on notification information **AY110** and **AY210**. This achieves information notification that creates a lesser sense of discomfort.

[0249] In addition, controller **A500** may adjust the chromaticity of each of notification information **AY110** and **AY210** between first display device **A100** and second display device **A200**, on the basis of the user surrounding information obtained from second imager **A600b**. More specifically, controller **A500** performs image processing on the image captured by second imager **A600b** to detect the viewpoint position of the driver, and obtains such viewpoint position as the user surrounding information. Controller **A500** estimates the chromaticity of the background when the driver is viewing first virtual image **A101**, on the basis of the viewpoint position. Controller **A500** reflects such estimation result on the chromaticity adjustment performed on notification information **AY110** and **AY210**. This achieves information notification that creates a lesser sense of discomfort.

#### Embodiment 12

[0250] The following describes Embodiment 12. FIG. 26 is a schematic diagram showing display system **A10C** according to Embodiment 12. As shown in FIG. 26, display system **A10C** further includes illuminance sensor **A700c** that detects the illuminance of an area in the vicinity of second display device **A200**. More specifically, illuminance sensor **A700c** detects the illuminance of the area in the vicinity of the output port of second display device **A200** from which video light is

outputted. Controller A500 adjusts the luminance of each of notification information AY110 and AY210 between first display device A100 and second display device A200, on the basis of the detection result of illuminance sensor A700c. When the illuminance of the area in the vicinity of the output port of second display device A200 is high due to direct sunlight coming through windshield A2, for example, the luminance of notification information AY210 is increased such that the visibility of notification information AY210 will not decrease. Then, the luminance of notification information AY110 is adjusted accordingly. With this, it is possible to reflect the illuminance of the area in the vicinity of second display device A200 on the luminance of each of notification information AY110 and AY210. This enables information notification that creates a lesser sense of discomfort.

### Embodiment 13

[0251] The following describes Embodiment 13. FIG. 27 is an explanatory diagram showing an example of displaying notification information AY110 and AY210 according to Embodiment 13. Embodiment 9 has shown an example case where display reference point APS is set on center line ALc in the width direction of first display area A102 due to the reason that target object AP is present outside first display area A102. Embodiment 13 describes the case where the target object AP is present inside first display area A102.

[0252] In FIG. 27, the size of first display area A102d is set to be greater in the vertical direction and the width direction than the size of first display area A102 of Embodiment 9, but may be the same size. Controller A500 performs image processing on the image data captured by first imager A400 to recognize at least one target object included in such image data and detects the position coordinates, the size, etc. of such target object. Controller A500 extracts target object AP that is present inside first display area A102d when seen from the viewpoint of the driver (reference eyepoint). Controller A500 then generates notification information AY110 and AY210, on the basis of the position coordinates, the size, etc. of such target object AP extracted, and causes first display device A100 and second display device A200 to display notification information AY110 and AY210. In so doing, since target object AP is present inside first display area A102d, controller A500 sets such target object AP as display reference point APS. Controller A500 causes notification information AY110 and AY210 to be displayed on straight line AL that connects the position of the coordinates of target object AP inside second display area A202 and display reference point APS.

[0253] As described above, when target object AP is present inside first display area A102d, notification information AY110 and AY210 are displayed, with target object AP serving as display reference point APS. This enables the driver to also recognize actual target object AP simply by viewing notification information AY110 and AY210.

[0254] Also, as shown in FIG. 27, controller A500 also causes mark AM to be displayed in the vicinity of display reference point APS inside first display area A102d. This mark AM may be a mark for alerting or a mark for warning. Such mark AM enables the driver to more easily notice the occurrence of notification information AY110. The driver can freely change the display content, the display period, etc. of mark AM. Also, mark AM is not limited to being displayed in an upper portion of target object AP, and thus may be displayed anywhere.

[0255] FIG. 28 is an explanatory diagram showing another example of displaying mark AM according to Embodiment 13. As shown in FIG. 28, second display area A202d may be divided into two parts, and the video captured by first imager A400, for example, may be displayed in one part, and mark AM may be displayed in the other part. In FIG. 28, frame AQ is displayed for target object AP to display target object AP in an emphasized manner. This emphasized display enables the driver to more easily notice target object AP.

[0256] FIG. 29 is an explanatory diagram showing another example of displaying notification information AY110 and AY210 according to Embodiment 13. FIG. 29 shows the case where target object AP is squatting down in front of vehicle A1, for example, when the vehicle is about to start,

and is invisible from the driver, although such target object AP is present within first display area **A102**. In this case, too, controller **A500** performs image processing on the image data captured by first imager **A400** to recognize target object AP, and detects the position coordinates, the size, etc. of such target object AP. Controller **A500** extracts target object AP, which is hidden behind hood **A3** when viewed from the viewpoint of the driver (reference eyepoint), but is present within first display area **A102**. Controller **A500** generates notification information **AY110** and **AY210**, on the basis of, for example, the position coordinates and the size of such target object AP extracted, and causes first display device **A100** and second display device **A200** to display notification information **AY110** and **AY210**. In so doing, since target object AP is present inside first display area **A102**, controller **A500** sets such target object AP as display reference point APS. Controller **A500** causes notification information **AY110** and **AY210** to be displayed on straight line AL that connects the coordinate position of target object AP inside second display area **A202** and display reference point APS.

[0257] As described above, even when target object AP that is invisible from the driver but is included in first display area **A102** is present, it is possible to enable the driver to recognize target object AP by guiding the line of sight of the driver by means of notification information **AY110** and **AY210**.

#### Embodiment 14

[0258] The following describes Embodiment 14. FIG. **30** is an explanatory diagram showing an example of displaying notification information **AY110e** and **AY210e** according to Embodiment 14. In Embodiment 14, controller **A500** causes notification information **AY110e** displayed by first display device **A100** and notification information **AY210e** displayed by second display device **A200** to be displayed in a continuous manner. More specifically, as shown in FIG. **30**, notification information **AY110e** and **AY210e** are represented in the form of a single arrow that spans first display area **A102** and second display area **A202**. As described above, since notification information **AY110e** and **AY210e** are displayed in a continuous manner, it is possible to enable the driver to more easily recognize the positional relationship between notification information **AY110e** and **AY210e**.

#### Embodiment 15

[0259] The following describes Embodiment 15. FIG. **31** is an explanatory diagram showing an example of displaying notification information **AY110f** and **AY210f** according to Embodiment 15. In Embodiment 15, controller **A500** causes notification information **AY110f** displayed by first display device **A100** and notification information **AY210f** displayed by second display device **A200** to be displayed in a manner that these items of notification information move from first display area **A102** to second display area **A202**.

[0260] For example, in the example shown in FIG. **31**, notification information **AY110f** is first displayed in first display area **A102** from display reference point APS along straight line AL, as shown in (a) in FIG. **31**. Subsequently, notification information **AY110f** moves along straight line AL toward second display area **A202**, as shown in (b) in FIG. **31**. Notification information **AY110f** and **AY210f** are then displayed, spanning first display area **A102** and second display area **A202** along straight line AL, as shown in (c) in FIG. **31**. At this time, notification information **AY110f** and **AY210f** are represented in the form a single continuous arrow.

[0261] As described above, notification information **AY110f** displayed by first display device **A100** and notification information **AY210f** displayed by second display device **A200** are displayed in a manner that these items of notification information move from first display area **A102** to second display area **A202**. This makes it easy to guide the line of sight of the driver.

[0262] FIG. **32** is an explanatory diagram showing another example of displaying notification information **AY110f** and **AY210f** according to Embodiment 15. In the example shown in FIG. **32**, notification information **AY110f** and **AY210f** are represented in the form of a plurality of triangles that are aligned on straight line AL. Notification information **AY110f** and **AY210f** are displayed in a

manner that these items of notification information gradually move from first display area **A102** to second display area **A202** (transit from the state shown in (a) in FIG. **32** to the state shown in (b) in FIG. **32**).

#### Embodiment 16

[0263] The following describes Embodiment 16. FIG. **33** is a schematic diagram showing display system **A10G** according to Embodiment 16. As shown in FIG. **33**, display system **A10G** further includes third imager **A650g** that captures images of lateral sides of vehicle **A1**, and tilt sensor **A670g** that detects a tilt of vehicle **A1**. Third imager **A650g** is provided, for example, on each of the left-side and right-side mirrors. Third imager **A650g** is an example of the side imager that captures images of the left side and the right side of vehicle **A1**. Controller **A500** controls second display device **A200** to cause it to display, in second display area **A202**, the video captured by third imager **A650g**.

[0264] FIG. **34** is an explanatory diagram showing an example of the display performed by second display device **A200** according to Embodiment 16. As shown in FIG. **34**, controller **A500** causes second display device **A200** to display pseudo image **AG10**, which mimics the interior of vehicle **A1**, as a blind spot portion of the driver who is seeing a lateral side of vehicle **A1**, in a manner that such pseudo image **AG10** is overlaid on video **AG20** captured by third imager **A650g**. Pseudo image **AG10** is represented in the form of a transparent image (represented with dot hatching in FIG. **34**) for the driver to be able to view video **AG20**. With this, it is possible for the driver to view pseudo image **AG10** and video **AG20** displayed in second display area **A202** of second display device **A200**. This enables the driver to recognize the situations of the lateral sides of vehicle **A1** and blind spots, and have a grasp of the surrounding situation of vehicle **A1**.

[0265] Controller **A500** may also correct video **AG20** captured by third imager **A650g**, on the basis of the detection result of tilt sensor **A670g**, and cause second display device **A200** to display the corrected video **AG20**. FIG. **35** is an explanatory diagram showing an example of displaying video **AG20** according to Embodiment 16. In FIG. **35**, the black frame indicates imaging range **Ag30** of third imager **A650g**, and the dashed line frame indicates display range **Ag40** displayed in second display area **A202** of second display device **A200**. When tilt sensor **A670g** detects a tilt of vehicle **A1** in the roll direction, for example, controller **A500** causes display range **Ag40** to move in the up-down direction within imaging range **Ag30** (see arrow **AYg1**), on the basis of such detection result, and corrects video **AG20**. Also, when tilt sensor **A670g** detects a tilt of vehicle **A1** in the pitch direction, controller **A500** causes display range **Ag40** to rotate within imaging range **Ag30** (see arrow **AYg2**), on the basis of such detection result, and corrects video **AG20**. As described above, since video **AG20** that has been corrected on the basis of the detection result of tilt sensor **A670g** is displayed by second display device **A200**, it is possible for the driver to have a more accurate grasp of the surrounding situation of vehicle **A1**.

[0266] Furthermore, controller **A500** may perform image processing on video **AG20** captured by third imager **A650g** to detect target object **AP1**. In this case, controller **A500** may cause target object **AP1** to be displayed in an emphasized manner.

[0267] FIG. **36** is an explanatory diagram showing another example of the display according to Embodiment 16. As shown in FIG. **36**, controller **A500** may control second display device **A200** to cause it to display frame **AQ1** for target object **AP1** inside second display area **A202**, thereby displaying target object **AP1** in an emphasized manner. This emphasized display enables the driver to easily notice target object **AP1**. Note that FIG. **36** omits the illustration of pseudo image **AG10** shown in second display area **A202**.

#### Embodiment 17

[0268] The following describes Embodiment 17. FIG. **37** is a schematic diagram showing display system **A10H** according to Embodiment 17. As shown in FIG. **37**, display system **A10H** further includes generator **A690h** that generates a sound or a vibration. When generator **A690h** is intended for generating a sound, examples of generator **A690h** include a speaker. When generator **A690h** is

intended for generating a vibration, examples of generator **A690h** include a vibration generator that causes the steering wheel to vibrate and a vibration generator that causes the driver's seat to vibrate. Controller **A500** controls generator **A690h** to cause it to generate the sound or the vibration, on the basis of the display timing of notification information **AY110** and **AY210**. With this, it is possible for the driver to more reliably have a grasp of notification information **AY110** and **AY210** in response to the sound or the vibration from generator **A690h**.

#### Embodiment 18

[0269] The following describes Embodiment 18. FIG. **38** is a schematic diagram showing display system **A10J** according to Embodiment 18. As shown in FIG. **38**, first display device **A100j** of display system **A10J** includes AR-HUD **A110j** and HUD **A120j**. Stated differently, first virtual image **A101j** displayed by first display device **A100j** is formed by AR-HUD **A110j** and HUD **A120j**. This enables more diverse representation of first virtual image **A101j**.

[0270] More specifically, AR-HUD **A110j** displays one portion **A111j** of first virtual image **A101j**. HUD **A120j** displays other portion **A121j** of first virtual image **A101j**. One portion **A111j** of first virtual image **A101j** is located at an angle such that one portion **A111j** is located upward, toward the front of vehicle **A1**. Other portion **A121j** of first virtual image **A101j** is located below such one portion **A111j** to be parallel to the up-down direction of vehicle **A1**.

[0271] FIG. **39** is an explanatory diagram showing an example of displaying notification information **AY110j** according to Embodiment 18. In FIG. **39**, first display area **A102j** is vertically divided into two parts, and upper area **A103j** corresponds to one portion **A111j** of first virtual image **A101j**, and lower area **A104j** corresponds to other portion **A121j** of first virtual image **A101j**. In FIG. **39**, navigation graphics **AY20j** is displayed in one portion **A111j** of first virtual image **A101j**. Also, notification information **AY110j** is displayed in other portion **A121j** of first virtual image **A101j**.

[0272] FIG. **40** and FIG. **41** are explanatory diagrams showing other examples of displaying notification information **AY110j** according to Embodiment 18. As shown in FIG. **40**, notification information **AY110j** displayed by first display device **A100j** may be displayed in a divided manner in upper area **A103j** and lower area **A104j**. In this case, since target object **AP** is present outside first display area **A102j**, display reference point **APS** is set on center line **ALc** in the width direction (width direction of the vehicle) in upper area **A103j**.

[0273] In the case shown in FIG. **41**, since target object **AP** is present inside lower area **A104j**, notification information **AY110j** may be displayed in lower area **A104j**, with such target object **AP** serving as display reference point **APS**. Note that in FIG. **41**, mark **AM** is shown in upper area **A103j**. As described above, depending on the position of target object **AP**, mark **AM** and **AY110j** may be displayed in a divided manner in upper area **A103j** and lower area **A104j**.

[0274] FIG. **42** is a schematic diagram showing another example of display system **A10J** according to Embodiment 18. As shown in FIG. **42**, AR-HUD **A110j** and HUD **A120j** may be integrally configured. Stated differently, one portion **A111j** of first virtual image **A101j** and other portion **A121j** of first virtual image **A101j** may be projected from a single device.

#### Embodiment 19

[0275] The following describes Embodiment 19. FIG. **43** is a schematic diagram showing display system **A10K** according to Embodiment 19. As shown in FIG. **43**, first virtual image **A101k** may be inclined in an orientation in which the upper end portion is located forward in the traveling direction of vehicle **A1** and the lower end portion is located rearward in the traveling direction of vehicle **A1**. Stated differently, the entirety of first virtual image **A101k** may be an image of AR-HUD **A110j** described in Embodiment 18. This enables first virtual image **A101k** to display video that creates a sense of depth in the eyes of the driver in the forward direction of vehicle **A1**. Since it is possible to display the notification information from the back toward the front, the driver's visibility is improved and a sense of discomfort is further reduced.

#### Embodiment 20



[0276] The following describes Embodiment 20. Embodiment 20 describes a variation of the first display device and the second display device.

[0277] FIG. 44 is a schematic diagram showing display system A10M according to Embodiment 20. As shown in FIG. 44, first display device A100m is a transmissive display laminated on windshield A2 and displays first image A101m. The transmissive display is, for example, a transparent organic EL display. Second display device A200m is a liquid crystal display and displays second image A201m. As described above, first display device A100m and second display device A200m may be display devices that do not project virtual images.

[0278] FIG. 45 is a schematic diagram showing display system A10N according to Embodiment 20. As shown in FIG. 45, first display device A100n is a transmissive display that is disposed closer to the interior side than windshield A2 is, and displays first image A101n. Second display device A200n is a liquid crystal display and displays second image A201n. As described above, first display device A100n and second display device A200n may be display devices that do not project virtual images.

[0279] FIG. 46 is a schematic diagram showing display system A10P according to Embodiment 20. As shown in FIG. 46, first display device A100p is a HUD, and projects first virtual image A101P ahead of vehicle A1. Second display device A200p is a liquid crystal display that displays second image A201p.

[0280] Note that, the configuration of FIG. 46 may be reversed, and the first display device may be a display that does not project virtual images, and the second display device may be a display that displays virtual images. Sated differently, the first display device and the second display device may be displays, at least one of which projects virtual images, or both of which do not project virtual images.

(Others)

[0281] The display system according to one or more aspects of the present disclosure has been described above on the basis of the embodiments, but the present disclosure is not limited to these embodiments. The scope of one or more aspects of the present disclosure may also include an embodiment achieved by making various modifications to the embodiments that can be conceived by those skilled in the art and an embodiment achieved by freely combining some of the elements in different embodiments without departing from the essence of the present disclosure.

[0282] For example, the foregoing embodiments have shown examples in which the imagers that image the surroundings of vehicle A1 are first imager A400, first imager A400a, and third imager A650g. However, an imager that captures an image of the rear of vehicle A1, an all-sky imager attached to the ceiling of vehicle A1, and a wide-angle imager may also be used as the imagers. Two or more of these imagers may be used to switch videos, depending on the direction in which target object AP is located. With this, it is possible to display a clearer video.

(Added Note)

[0283] From the embodiments and so forth described above, techniques described below are disclosed.

[Technique 1]

[0284] A display system including: [0285] a first display device that projects a first virtual image ahead of a user in a vehicle; and [0286] a second display device that projects a second virtual image ahead of the user and below the first virtual image, [0287] wherein a lower end of the first virtual image and an upper end of the second virtual image are located within a circle whose diameter is a length of the second virtual image in a side view.

[Technique 2]

[0288] The display system according to Technique 1, [0289] wherein a difference between a first angle of depression and a second angle of depression is 12° or less, the first angle of depression being an angle formed by the lower end of the first virtual image with respect to a viewpoint of the user, the second angle of depression being an angle formed by the upper end of the second virtual

image with respect to the viewpoint of the user.

[Technique 3]

[0290] The display system according to Technique 1 or 2, including: [0291] a third display device that projects a third virtual image ahead of the user, [0292] wherein the third virtual image is located at a viewing distance of 0.25 diopters or less with respect to a viewing distance of the second virtual image.

[Technique 4]

[0293] The display system according to any one of Techniques 1 to 3, [0294] wherein at least one of the first virtual image projected by the first display device or the second virtual image projected by the second display device is displayed in a plurality of layers.

[Technique 5]

[0295] The display system according to any one of Techniques 1 to 4, [0296] wherein the second display device includes: [0297] a display element that emits video light for forming the second virtual image; and [0298] an optical system for projecting the video light from the display element as the second virtual image, and [0299] the optical system includes a concave mirror that reflects the video light at a final stage.

[Technique 6]

[0300] The display system according to any one of Techniques 1 to 5, [0301] wherein at least one of the first display device or the second display device includes an adjuster for adjusting the viewing distance of the first virtual image and the viewing distance of the second virtual image.

[Technique 7]

[0302] The display system according to any one of Techniques 1 to 6, including: [0303] a controller that controls a display content of the first display device and a display content of the second display device.

[Technique 8]

[0304] The display system according to any one of Techniques 1 to 7, [0305] wherein at least one of the first display device or the second display device includes an optical system that includes a hologram element.

[Technique 9]

[0306] The display system according to any one of Techniques 1 to 8, including: [0307] a first position adjuster that adjusts a first display position in an up-down direction of the first virtual image; [0308] a second position adjuster that adjusts a second display position in an up-down direction of the second virtual image; and [0309] a coordinate changer that changes coordinates of the first virtual image within a displayable range of the first virtual image, based on the first display position and the second display position that have been adjusted.

[Technique 10]

[0310] The display system according to Technique 9, [0311] wherein the first display device includes a first reflector that reflects video light for forming the first virtual image, [0312] the second display device includes a second reflector that reflects video light for forming the second virtual image, [0313] the first position adjuster adjusts at least one of an orientation or a position of the first reflector to adjust the first display position of the first virtual image, and [0314] the second position adjuster adjusts at least one of an orientation or a position of the second reflector to adjust the second display position of the second virtual image.

[Technique 11]

[0315] The display system according to claim **9** or **10**, including: [0316] a head imager that captures an image of a head of the user; [0317] an estimator that estimates a viewpoint position of the user, based on the image captured by the head imager; and [0318] an adjustment controller that controls the first position adjuster and the second position adjuster, based on the viewpoint position of the user estimated.

[Technique 12]

[0319] The display system according to any one of Techniques 1 to 11, including: [0320] a first imager that captures images of surroundings of the vehicle; and [0321] a controller that controls the first display device, the second display device, and the first imager, [0322] wherein when the controller causes each of the first display device and the second display device to display notification information to be provided to the user, the controller adjusts, between the first display device and the second display device, at least one of luminance, a size, or chromaticity of the notification information to be displayed together with vehicle surrounding information obtained from the first imager.

[Technique 13]

[0323] The display system according to Technique 12, [0324] wherein the first imager captures the images of an area ahead of the vehicle, and [0325] the controller calculates luminance of the first virtual image in a state of being overlaid on a background, based on the vehicle surrounding information, and adjusts the luminance of the notification information between the first display device and the second display device, based on a calculation result.

[Technique 14]

[0326] The display system according to Technique 12 or 13, [0327] wherein the controller adjusts the luminance of the notification information between the first display device and the second display device to cause the luminance of the notification information in the first virtual image and the luminance of the notification information in the second virtual image to be approximately same.

[Technique 15]

[0328] The display system according to Technique 12 or 13, [0329] wherein the controller adjusts the luminance of the notification information between the first display device and the second display device to cause the luminance of the notification information in the first virtual image to be higher than the luminance of the notification information in the second virtual image.

[Technique 16]

[0330] The display system according to Technique 12 or 13, [0331] wherein the controller adjusts the luminance of the notification information between the first display device and the second display device to cause the luminance of the notification information in the first virtual image to be lower than the luminance of the notification information in the second virtual image.

[Technique 17]

[0332] The display system according to any one of Techniques 12 to 16, including: [0333] a second imager that captures an image of surroundings of the user, [0334] wherein the controller adjusts the luminance of the notification information between the first display device and the second display device, based on user surrounding information obtained from the second imager.

[Technique 18]

[0335] The display system according to any one of Techniques 12 to 16, including: [0336] an illuminance sensor that detects an illuminance of an area in the vicinity of the second display device, [0337] wherein the controller adjusts the luminance of the notification information between the first display device and the second display device, based on a detection result of the illuminance sensor.

[Technique 19]

[0338] The display system according to any one of Techniques 12 to 18, including: [0339] a second imager that captures an image of surroundings of the user, [0340] wherein the controller adjusts the size of the notification information between the first display device and the second display device, based on user surrounding information obtained from the second imager.

[Technique 20]

[0341] The display system according to any one of Techniques 12 to 19, [0342] wherein the first imager captures the images of an area ahead of the vehicle, and [0343] the controller calculates chromaticity of the first virtual image in a state of being overlaid on a background, based on the vehicle surrounding information, and adjusts, between the first display device and the second

display device, the chromaticity of the notification information in the first virtual image and the chromaticity of the notification information in the second virtual image to be approximately same, based on a calculation result.

[Technique 21]

[0344] The display system according to any one of Techniques 12 to 19, [0345] wherein the controller adjusts the chromaticity of the notification information in the first virtual image, with the chromaticity of the notification information in the second virtual image kept constant.

[Technique 22]

[0346] The display system according to any one of Techniques 12 to 19, [0347] wherein the controller adjusts the chromaticity of the notification information in the second virtual image to be approximately same as the chromaticity of the notification information in the first virtual image, with the chromaticity of the notification information in the first virtual image kept constant.

[Technique 23]

[0348] The display system according to any one of Techniques 20 to 22, including: [0349] a second imager that captures an image of surroundings of the user, [0350] wherein the controller adjusts the chromaticity the notification information between the first display device and the second display device, based on user surrounding information obtained from the second imager.

[Technique 24]

[0351] The display system according to any one of Techniques 12 to 23, [0352] wherein the controller causes the notification information to be displayed, based on a display reference point that is set inside a first display area of the first display device, [0353] when a target object for which the notification information is provided is present inside the first display area, the controller sets the target object as the display reference point, and [0354] when the target object for which the notification information is provided is present outside the first display area, the controller sets the display reference point on a center line in a width direction of the first display area.

[Technique 25]

[0355] The display system according to Technique 24, [0356] wherein the controller causes the notification information displayed by the first display device and the notification information displayed by the second display device to be located on a straight line that includes the display reference point.

[Technique 26]

[0357] The display system according to Technique 25, [0358] wherein the controller causes the notification information displayed by the first display device and the notification information displayed by the second display device to be displayed in a continuous manner.

[Technique 27]

[0359] The display system according to Technique 25, [0360] wherein the controller causes the notification information displayed by the first display device and the notification information displayed by the second display device to be displayed in a manner that each of the notification information displayed by the first display device and the notification information displayed by the second display device moves from the first display area of the first display device toward a second display area of the second display device.

[Technique 28]

[0361] The display system according to any one of Techniques 12 to 27, [0362] wherein the controller causes the second display device to display video captured by the first imager.

[Technique 29]

[0363] The display system according to any one of Techniques 12 to 28, including: [0364] a third imager that captures video of surroundings of a lateral side of the vehicle, [0365] wherein the controller causes the second display device to display a pseudo image overlaid on the video captured by the third imager, the pseudo image being an image that mimics an interior of the vehicle as a blind spot of the user.

[Technique 30]

[0366] The display system according to Technique 29, including: [0367] a tilt sensor that obtains a tilt of the vehicle, [0368] wherein the controller corrects the video captured by the third imager, based on a detection result of the tilt sensor, and causes the second display device to display the video corrected.

[Technique 31]

[0369] The display system according to any one of Techniques 12 to 30, including: [0370] a generator that generates a sound or a vibration, [0371] wherein the controller causes the generator to generate the sound or the vibration, based on a display timing of the notification information.

[Technique 32]

[0372] The display system according to any one of Techniques 1 to 31, [0373] wherein the first display device includes: an augmented reality head-up display that projects one portion of the first virtual image; and a head-up display that projects another portion of the first virtual image.

[0374] While various embodiments have been described herein above, it is to be appreciated that various changes in form and detail may be made without departing from the spirit and scope of the present disclosure as presently or hereafter claimed.

Further Information about Technical Background to this Application

[0375] The disclosures of the following patent applications including specification, drawings, and claims are incorporated herein by reference in their entirety: Japanese Patent Application No. 2024-031098 filed on Mar. 1, 2024, Japanese Patent Application No. 2024-031110 filed on Mar. 1, 2024, and Japanese Patent Application No. 2024-154389 filed on Sep. 9, 2024.

INDUSTRIAL APPLICABILITY

[0376] The present disclosure is applicable for use as a display system for displaying images.

## Claims

1. A display system comprising: a first display device that projects a first virtual image ahead of a user in a vehicle; and a second display device that projects a second virtual image ahead of the user and below the first virtual image, wherein a lower end of the first virtual image and an upper end of the second virtual image are located within a circle whose diameter is a length of the second virtual image in a side view.
2. The display system according to claim 1, wherein a difference between a first angle of depression and a second angle of depression is  $12^\circ$  or less, the first angle of depression being an angle formed by the lower end of the first virtual image with respect to a viewpoint of the user, the second angle of depression being an angle formed by the upper end of the second virtual image with respect to the viewpoint of the user.
3. The display system according to claim 1, comprising: a third display device that projects a third virtual image ahead of the user, wherein the third virtual image is located at a viewing distance of 0.25 diopters or less with respect to a viewing distance of the second virtual image.
4. The display system according to claim 1, wherein at least one of the first virtual image projected by the first display device or the second virtual image projected by the second display device is displayed in a plurality of layers.
5. The display system according to claim 1, wherein the second display device includes: a display element that emits video light for forming the second virtual image; and an optical system for projecting the video light from the display element as the second virtual image, and the optical system includes a concave mirror that reflects the video light at a final stage.
6. The display system according to claim 1, comprising: a first position adjuster that adjusts a first display position in an up-down direction of the first virtual image; a second position adjuster that adjusts a second display position in an up-down direction of the second virtual image; and a coordinate changer that changes coordinates of the first virtual image within a displayable range of

the first virtual image, based on the first display position and the second display position that have been adjusted.

**7.** The display system according to claim 6, wherein the first display device includes a first reflector that reflects video light for forming the first virtual image, the second display device includes a second reflector that reflects video light for forming the second virtual image, the first position adjuster adjusts at least one of an orientation or a position of the first reflector to adjust the first display position of the first virtual image, and the second position adjuster adjusts at least one of an orientation or a position of the second reflector to adjust the second display position of the second virtual image.

**8.** The display system according to claim 6, comprising: a head imager that captures an image of a head of the user; an estimator that estimates a viewpoint position of the user, based on the image captured by the head imager; and an adjustment controller that controls the first position adjuster and the second position adjuster, based on the viewpoint position of the user estimated.

**9.** The display system according to claim 1, comprising: a first imager that captures images of surroundings of the vehicle; and a controller that controls the first display device, the second display device, and the first imager, wherein when the controller causes each of the first display device and the second display device to display notification information to be provided to the user, the controller adjusts, between the first display device and the second display device, at least one of luminance, a size, or chromaticity of the notification information to be displayed together with vehicle surrounding information obtained from the first imager.

**10.** The display system according to claim 9, wherein the first imager captures the images of an area ahead of the vehicle, and the controller calculates luminance of the first virtual image in a state of being overlaid on a background, based on the vehicle surrounding information, and adjusts the luminance of the notification information between the first display device and the second display device, based on a calculation result.

**11.** The display system according to claim 9, wherein the controller adjusts the luminance of the notification information between the first display device and the second display device to cause the luminance of the notification information in the first virtual image and the luminance of the notification information in the second virtual image to be approximately same.

**12.** The display system according to claim 9, wherein the first imager captures the images of an area ahead of the vehicle, and the controller calculates chromaticity of the first virtual image in a state of being overlaid on a background, based on the vehicle surrounding information, and adjusts, between the first display device and the second display device, the chromaticity of the notification information in the first virtual image and the chromaticity of the notification information in the second virtual image to be approximately same, based on a calculation result.

**13.** The display system according to claim 9, wherein the controller adjusts the chromaticity of the notification information in the first virtual image, with the chromaticity of the notification information in the second virtual image kept constant.

**14.** The display system according to claim 9, wherein the controller causes the notification information to be displayed, based on a display reference point that is set inside a first display area of the first display device, when a target object for which the notification information is provided is present inside the first display area, the controller sets the target object as the display reference point, and when the target object for which the notification information is provided is present outside the first display area, the controller sets the display reference point on a center line in a width direction of the first display area.

**15.** The display system according to claim 14, wherein the controller causes the notification information displayed by the first display device and the notification information displayed by the second display device to be located on a straight line that includes the display reference point.

**16.** The display system according to claim 15, wherein the controller causes the notification information displayed by the first display device and the notification information displayed by the

second display device to be displayed in a continuous manner.

**17.** The display system according to claim 15, wherein the controller causes the notification information displayed by the first display device and the notification information displayed by the second display device to be displayed in a manner that each of the notification information displayed by the first display device and the notification information displayed by the second display device moves from the first display area of the first display device toward a second display area of the second display device.

**18.** The display system according to claim 9, wherein the controller causes the second display device to display video captured by the first imager.

**19.** The display system according to claim 9, comprising: a side imager that captures video of surroundings of a lateral side of the vehicle, wherein the controller causes the second display device to display a pseudo image overlaid on the video captured by the side imager, the pseudo image being an image that mimics an interior of the vehicle as a blind spot of the user.

**20.** The display system according to claim 9, comprising: a generator that generates a sound or a vibration, wherein the controller causes the generator to generate the sound or the vibration, based on a display timing of the notification information.

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